Bayesian Learning - Computer Lab 3

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1. Gibbs sampling for the logistic regression

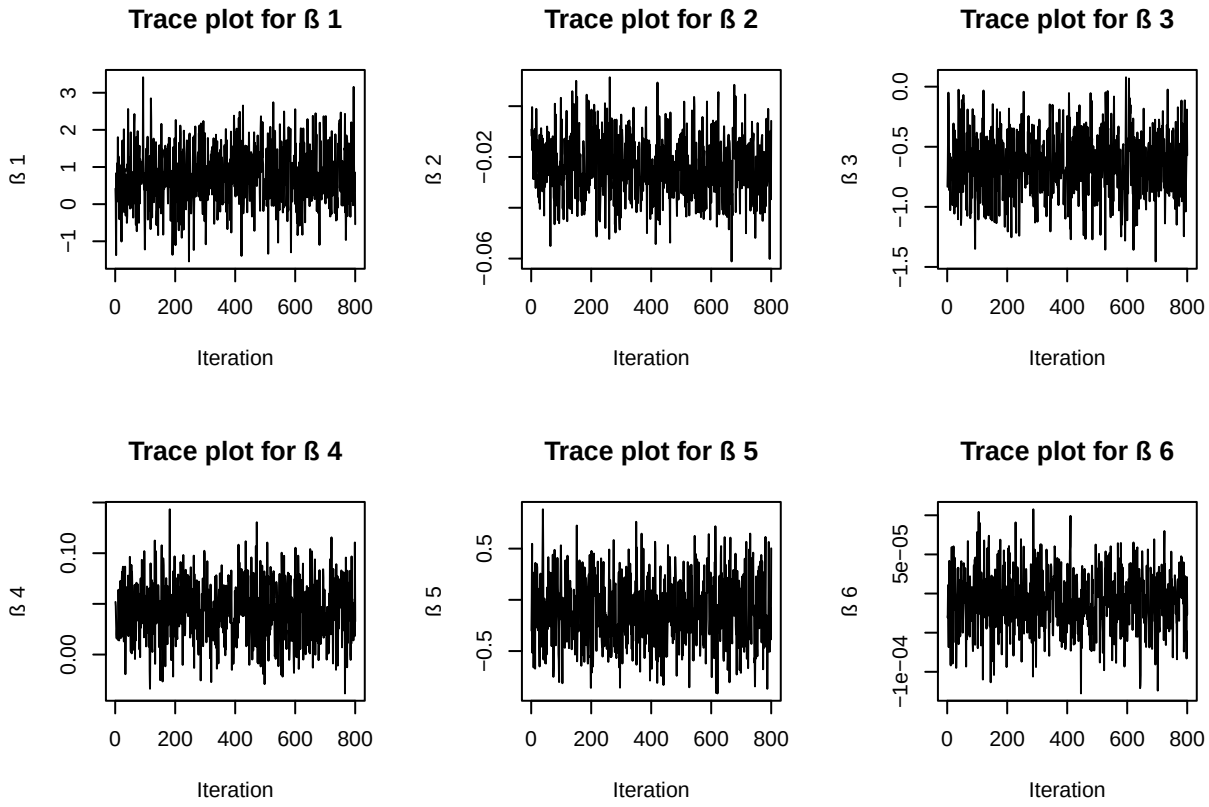
Consider again the logistic regression model in problem 2 from the previous computer lab 2. Use the prior $\beta \sim \mathcal{N}(0, \tau^2 I)$, where $\tau = 3$.

(a)

Implement (code!) a Gibbs sampler that simulates from the joint posterior $p(\omega, \beta|x)$ by augmenting the data with Polya-gamma latent variables $\omega_i, i = 1, \ldots, n$. The full conditional posteriors are given on the slides from Lecture 7. Evaluate the convergence of the Gibbs sampler by calculating the Inefficiency Factors (IFs) and by plotting the trajectories of the sampled Markov chains.

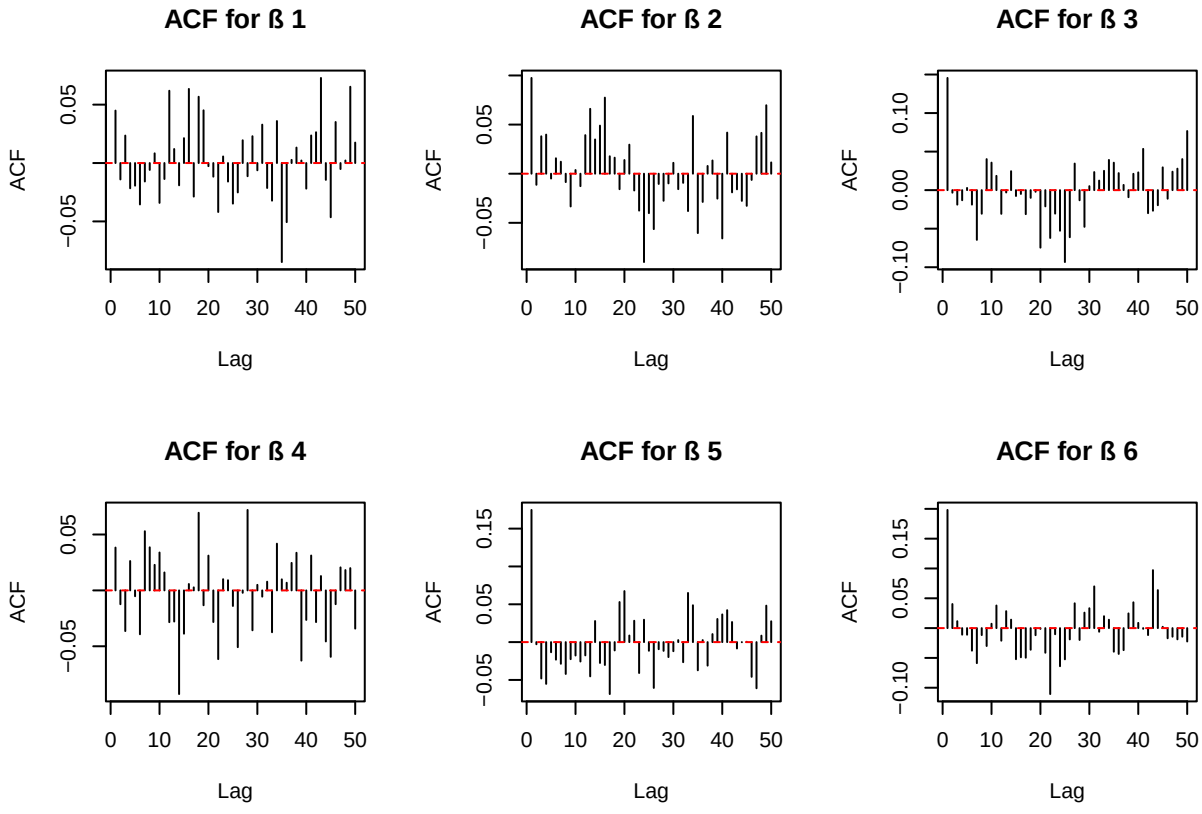
```
library(BayesLogit)
library(mvtnorm)

data = read.csv('Disease.csv')
x <- as.matrix(data [, 1:6])
y <- as.matrix(data [, 7])
# initialize beta
nDraws = 1000
burnin <- 200       # Burn-in period

n <- nrow(x)
p <- ncol(x)

beta <- rep(1, 6) # 5+1 intercept

beta_samples <- matrix(NA, nDraws, p)

set.seed(123)

for (i in 1:nDraws) {
  omega <- rpg(n, 1, x %*% beta)
  k <- y - 1 / 2
  B <- 9 * diag(6)
  b <- rep(0, p)
  # Conditional posterior for beta

  v_omg <- solve(t(x) %*% diag(omega) %*% x + solve(B))
  m_omg <- v_omg %*% (t(x) %*% k + solve(B) %*% b)
```

```r
  beta <- as.numeric(rmvnorm(1, mean = m_omg, sigma = v_omg))
  beta_samples[i, ] <- beta
}
# Discard burn-in
beta_samples <- beta_samples[(burnin + 1):nDraws, ]
nPost <- nrow(beta_samples)

par(mfrow = c(2, 3))  # 6 pics

for (i in 1:p) {
  plot(
    beta_samples[, i],
    type = 'l',
    main = paste("Trace plot for  ", i),
    xlab = "Iteration",
    ylab = paste(" ", i)
  )
}
```




```r
max_lag <- 50  # set the max lag
acfs <- matrix(NA, max_lag, p)

for (i in 1:p) {
  chain <- beta_samples[, i] - mean(beta_samples[, i])
  var_chain <- mean(chain ^ 2)
```

```r
  for (j in 1:max_lag) {  # j as lag
    # calculate the autocorrelation factor
    acfs[j, i] <- sum(chain[1:(nPost - j)] * chain[(j + 1):nPost]) / nPost / var_chain

  }

}

par(mfrow = c(2, 3))  # 6 pics

for (i in 1:p) {
  plot(
    1:max_lag,
    acfs[, i],
    type = 'h',
    main = paste("ACF for  ", i),
    xlab = "Lag",
    ylab = "ACF"
  )
  abline(h = 0, col = "red", lty = 2)
}
```




```r
# inefficiency factor (IF)
IFs <- numeric(p)
```

```r
for (i in 1:p) {
  IFs[i] <- 1 + 2 * sum(acfs[, i])
  cat(paste("IF for  ", i, "=", round(IFs[i], 2), "\n"))
}
```

```
## IF for   1 = 1.16
## IF for   2 = 1.25
## IF for   3 = 0.95
## IF for   4 = 0.72
## IF for   5 = 0.77
## IF for   6 = 0.72
```

Plots show that all efficient fluctrate around certain levels without any obvious tendency or any stuck. And calculated inefficiency factors are around 1, which means very sampling are performed with quite low dependency and the efficiency is very good.

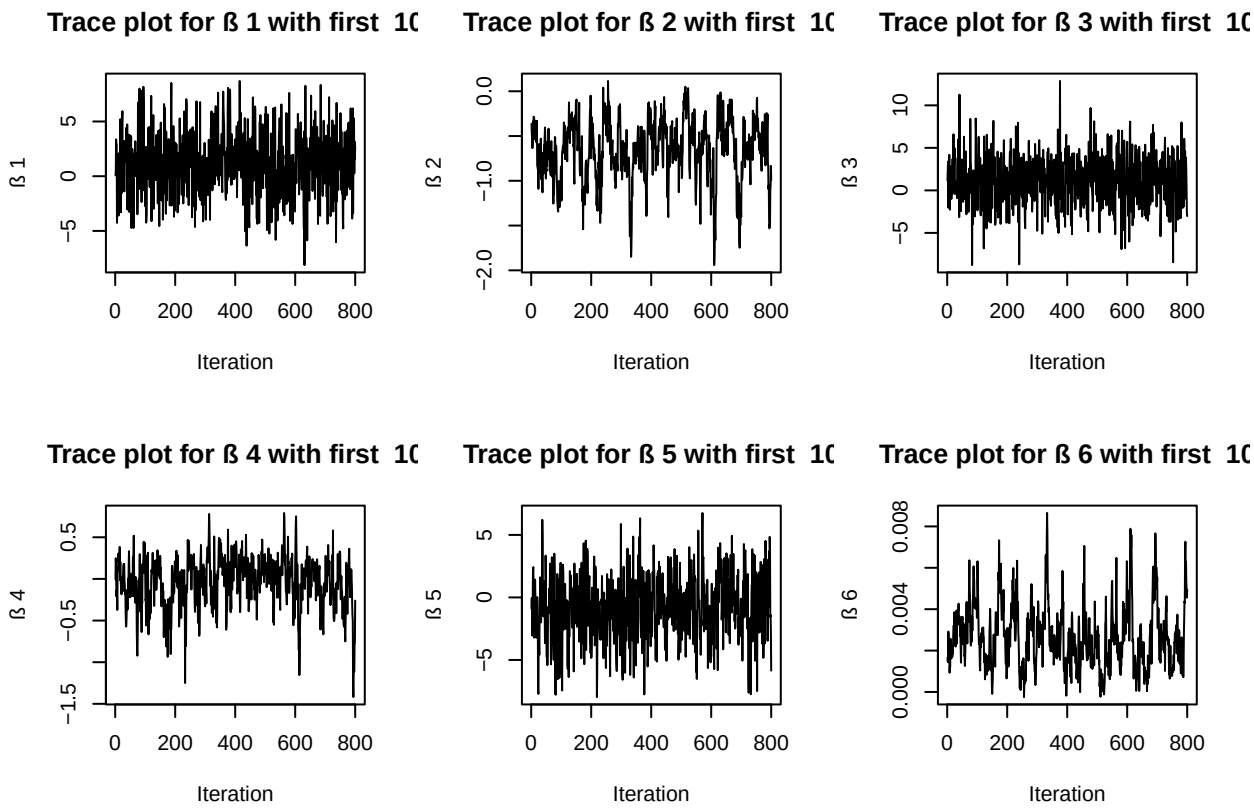
(b)

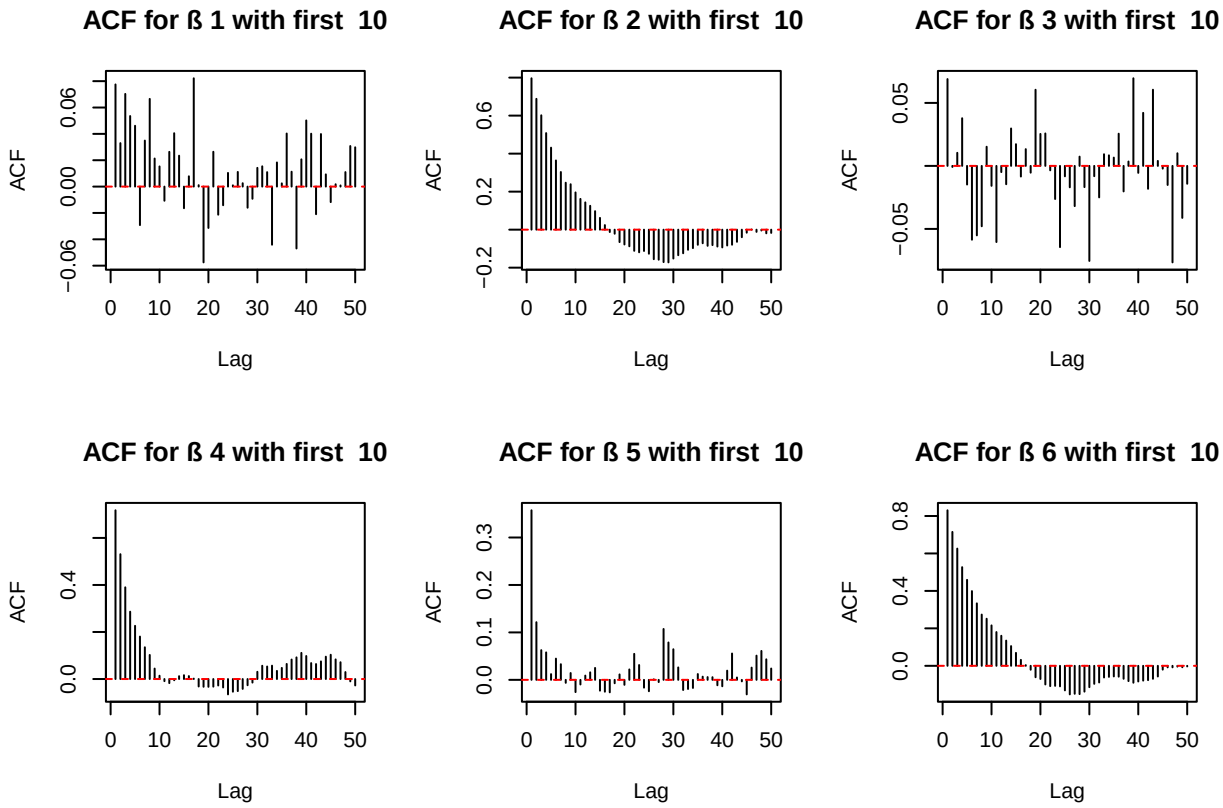
Repeat the same task as in (a), but now only use the first m observations of the dataset. Do this for all $m \in \{10, 40, 80\}$.

```r
m <- c(10, 40, 80)
for (z in 1:3) {
  row_nr <- m[z]
  data = read.csv('Disease.csv')
  x <- as.matrix(data [1:row_nr, 1:6])
  y <- as.matrix(data [1:row_nr, 7])

  # initialize beta
nDraws = 1000
burnin <- 200       # Burn-in period

n <- nrow(x)
p <- ncol(x)

beta <- rep(1, 6) # 5+1 intercept

beta_samples <- matrix(NA, nDraws, p)

set.seed(123)

for (i in 1:nDraws) {
  omega <- rpg(n, 1, x %*% beta)
  k <- y - 1 / 2
  B <- 9 * diag(6)
  b <- rep(0, p)
  # Conditional posterior for beta

  v_omg <- solve(t(x) %*% diag(omega) %*% x + solve(B))
  m_omg <- v_omg %*% (t(x) %*% k + solve(B) %*% b)
```

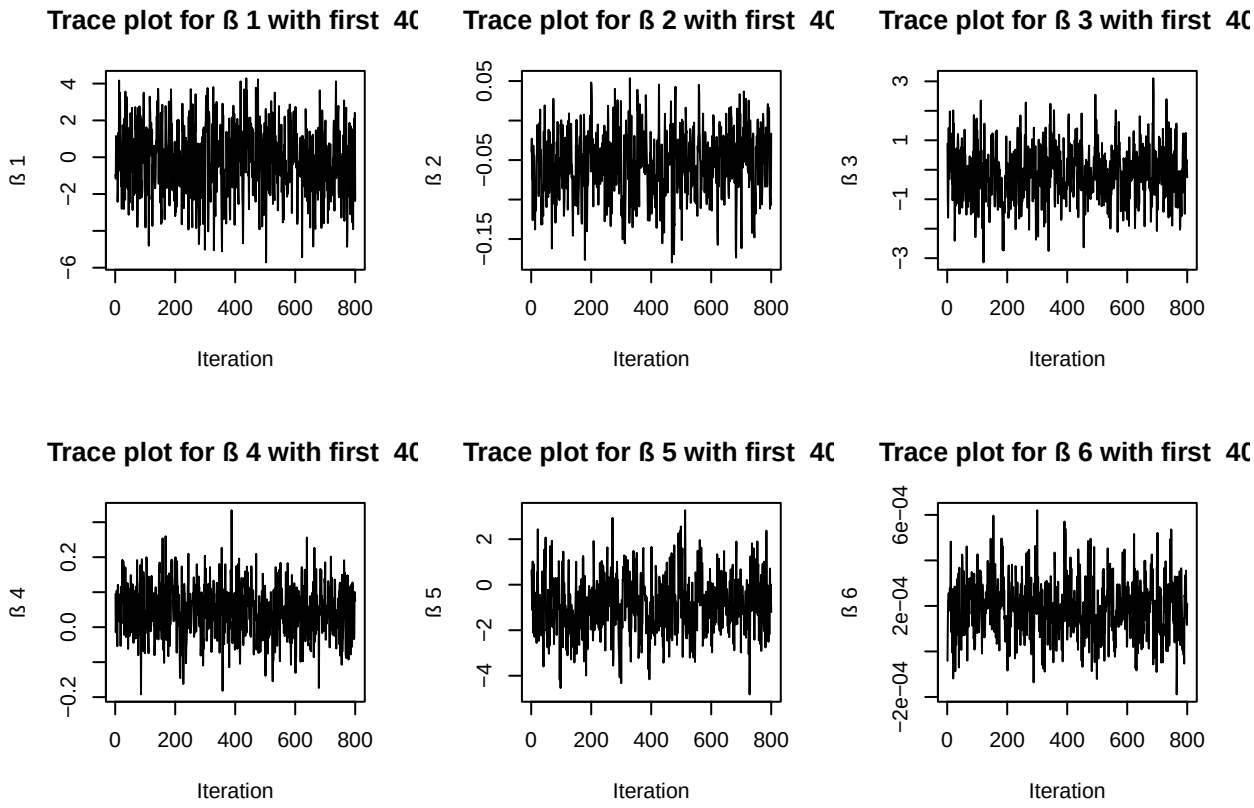
```
  beta <- as.numeric(rmvnorm(1, mean = m_omg, sigma = v_omg))
  beta_samples[i, ] <- beta
}
# Discard burn-in
beta_samples <- beta_samples[(burnin + 1):nDraws, ]
nPost <- nrow(beta_samples)

par(mfrow = c(2, 3))  # 6 pics

for (i in 1:p) {
  plot(
    beta_samples[, i],
    type = 'l',
    main = paste("Trace plot for  ", i , "with first ", row_nr),
    xlab = "Iteration",
    ylab = paste(" ", i)
  )
}


max_lag <- 50  # set the max lag
acfs <- matrix(NA, max_lag, p)

for (i in 1:p) {
  chain <- beta_samples[, i] - mean(beta_samples[, i])
  var_chain <- mean(chain ^ 2)
  for (j in 1:max_lag) {  # j as lag
    # calculate the autocorrelation factor
    acfs[j, i] <- sum(chain[1:(nPost - j)] * chain[(j + 1):nPost]) / nPost / var_chain

  }

}

par(mfrow = c(2, 3))  # 6 pics

for (i in 1:p) {
  plot(
    1:max_lag,
    acfs[, i],
    type = 'h',
    main = paste("ACF for  ", i , "with first ", row_nr),
    xlab = "Lag",
    ylab = "ACF"
  )
  abline(h = 0, col = "red", lty = 2)
}

# inefficiency factor (IF)
IFs <- numeric(p)

for (i in 1:p) {
  IFs[i] <- 1 + 2 * sum(acfs[, i])
```

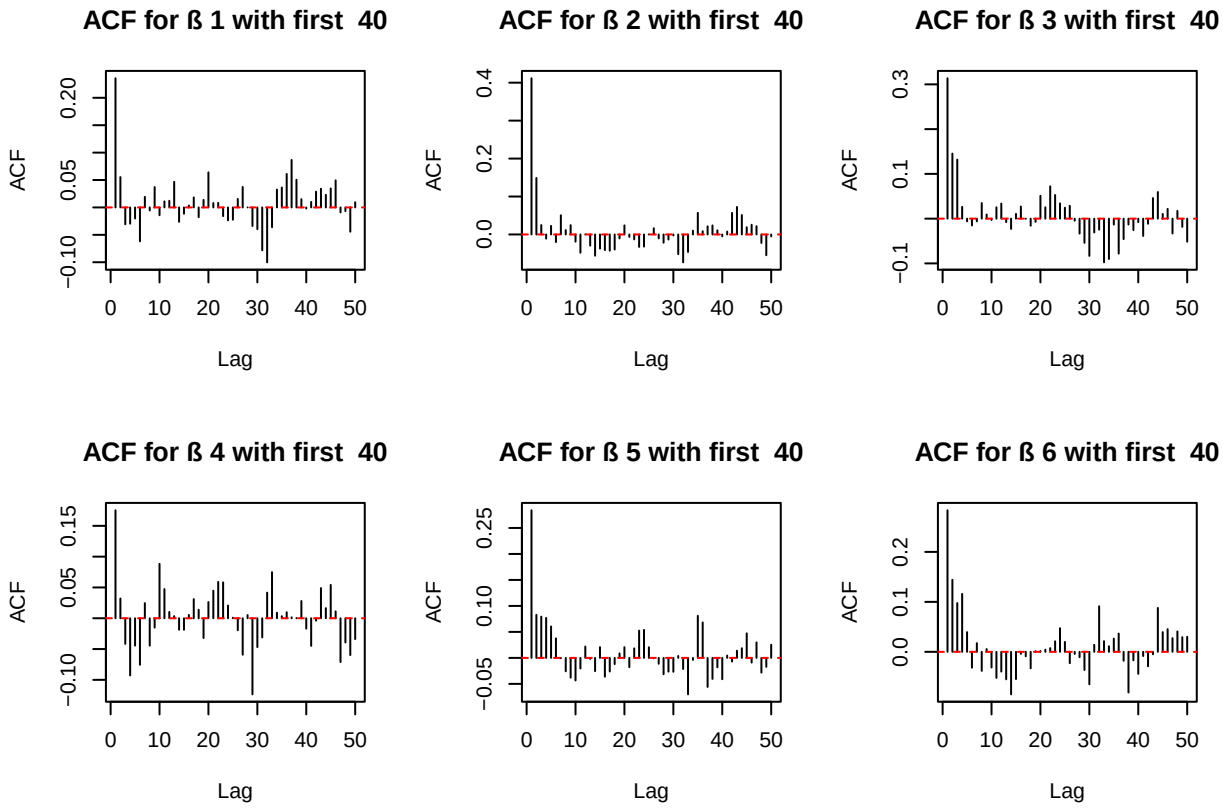
```
  cat(paste("IF for  ", i, "=", round(IFs[i], 2), "with first ", row_nr, "\n"))
}

}
```



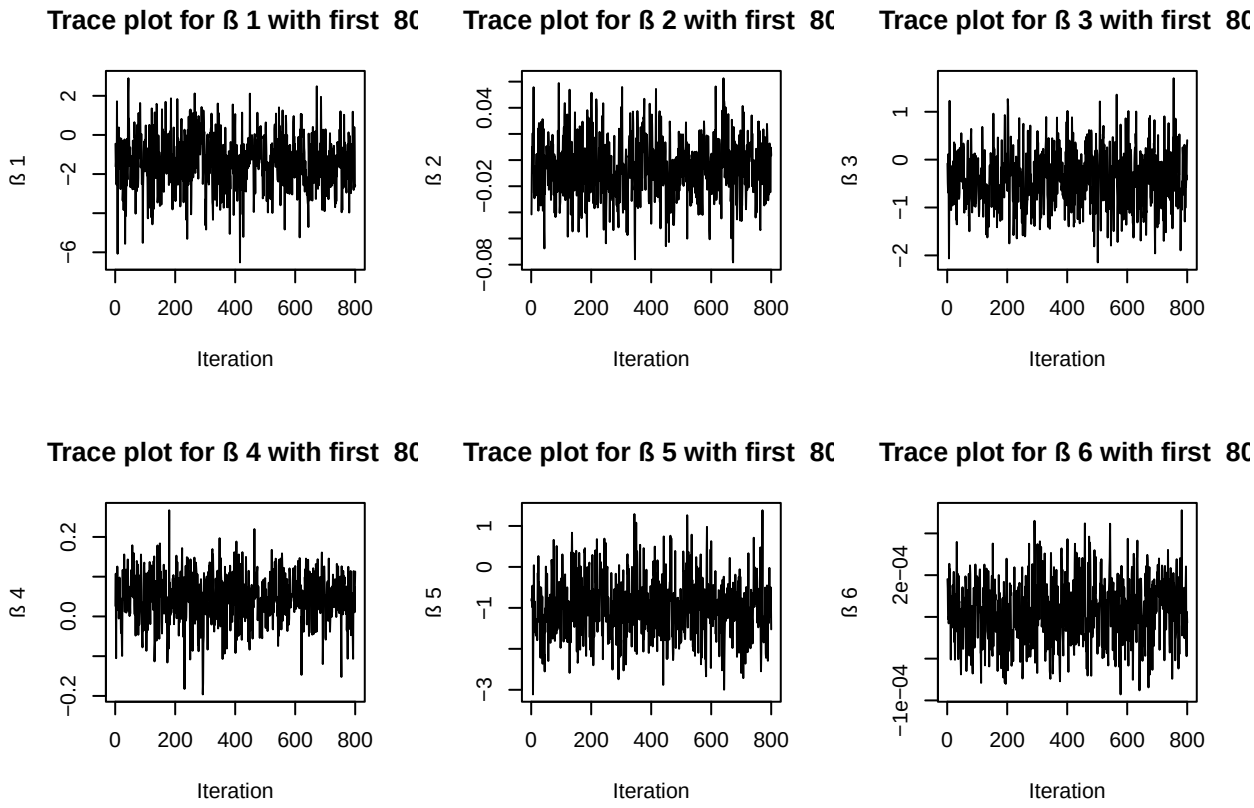


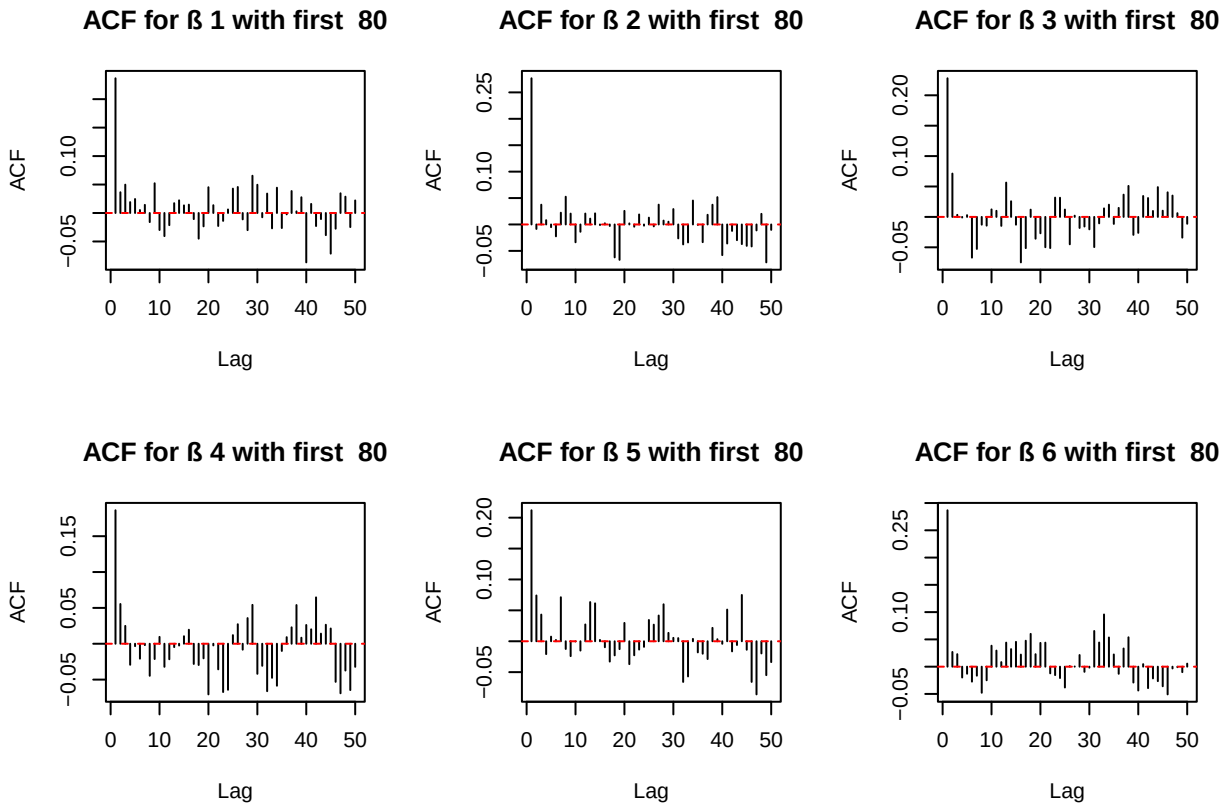

```
## IF for   1 = 2.34 with first  10
## IF for   2 = 5.3 with first  10
## IF for   3 = 0.59 with first  10
## IF for   4 = 7.94 with first  10
## IF for   5 = 3.34 with first  10
## IF for   6 = 6.65 with first  10
```


Trace plot for ß 1 with first 40
ß 1
−6
−2
0
2
4
0
200
400
600
800
Iteration
Trace plot for ß 2 with first 40
ß 2
−0.15
−0.05
0.05
0
200
400
600
800
Iteration
Trace plot for ß 3 with first 40
ß 3
−3
−1
1
3
0
200
400
600
800
Iteration
Trace plot for ß 4 with first 40
ß 4
−0.2
0.0
0.2
0
200
400
600
800
Iteration
Trace plot for ß 5 with first 40
ß 5
−4
−2
0
2
0
200
400
600
800
Iteration
Trace plot for ß 6 with first 40
ß 6
−2e−04
2e−04
6e−04
0
200
400
600
800
Iteration




```
## IF for   1 = 1.85 with first  40
## IF for   2 = 1.75 with first  40
## IF for   3 = 1.73 with first  40
## IF for   4 = 1.02 with first  40
## IF for   5 = 1.95 with first  40
## IF for   6 = 2.06 with first  40
```


Trace plot for ß 1 with first 80
ß 1
−6
−2
0
2
0
200
400
600
800
Iteration
Trace plot for ß 2 with first 80
ß 2
−0.08
−0.02
0.04
0
200
400
600
800
Iteration
Trace plot for ß 3 with first 80
ß 3
−2
−1
0
1
0
200
400
600
800
Iteration
Trace plot for ß 4 with first 80
ß 4
−0.2
0.0
0.2
0
200
400
600
800
Iteration
Trace plot for ß 5 with first 80
ß 5
−3
−1
0
1
0
200
400
600
800
Iteration
Trace plot for ß 6 with first 80
ß 6
−1e−04
2e−04
0
200
400
600
800
Iteration




```
## IF for  1 = 1.83 with first  80
## IF for  2 = 1.15 with first  80
## IF for  3 = 1.23 with first  80
## IF for  4 = 0.36 with first  80
## IF for  5 = 1.42 with first  80
## IF for  6 = 2.31 with first  80
```

As for case 1 where we have 10 observations, there is a decreasing tendency noticed for $\beta 4$ and the overall IF is comparatively high, thus it is not considered to converge.

For case 2 and 3 where there are 40 and 80 observations respectively, all plots have reached a stabilization and all parameters have a good IF factor. Thus convergence is thought to be reached.

---

2. Metropolis Random Walk for Poisson regression

Consider the following Poisson regression model:

$$y_i|\beta \sim \text{Poisson}\left(\exp(x_i^T\beta)\right), \quad i = 1, \ldots, n,$$

where y_i is the count for the ith observation in the sample and x_i is the p-dimensional vector with covariate observations for the ith observation.

The dataset contains observations from 700 eBay auctions of coins. The response variable is `nBids` and records the number of bids in each auction. The remaining variables are features/covariates (x):

- **Const**: Intercept
- **PowerSeller**: Equal to 1 if the seller is selling large volumes on eBay
- **VerifyID**: Equal to 1 if the seller is a verified seller by eBay
- **Sealed**: Equal to 1 if the coin was sold in an unopened envelope
- **MinBlem**: Equal to 1 if the coin has a minor defect
- **MajBlem**: Equal to 1 if the coin has a major defect
- **LargNeg**: Equal to 1 if the seller received a lot of negative feedback from customers
- **LogBook**: Logarithm of the book value of the auctioned coin according to expert sellers (Standardized)
- **MinBidShare**: Ratio of the minimum selling price (starting price) to the book value (Standardized).

(a)

Obtain the maximum likelihood estimator of β in the Poisson regression model for the eBay data. [Hint: `glm.R`, don't forget that `glm()` adds its own intercept so don't input the covariate `Const`.] Which covariates are significant?

```
rm(list=ls())

data <-read.table('eBayNumberOfBidderData_2025.dat', header = FALSE, skip = 1)

colnames(data) <- c("nBids", "Const", "PowerSeller", "VerifyID", "Sealed", "MinBlem", "MajBlem", "LargN

# Fit Poisson regression model (excluding Const)

model <- glm(nBids ~ . - Const, family = poisson, data = data)
summary(model)
```

```
##
## Call:
## glm(formula = nBids ~ . - Const, family = poisson, data = data)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  1.08534    0.03592  30.213  < 2e-16 ***
## PowerSeller -0.02833    0.04433  -0.639  0.52280
## VerifyID    -0.34902    0.13008  -2.683  0.00729 **
## Sealed       0.50101    0.06676   7.504 6.18e-14 ***
## MinBlem     -0.12189    0.08388  -1.453  0.14619
## MajBlem     -0.24884    0.09865  -2.523  0.01165 *
## LargNeg      0.03610    0.07075   0.510  0.60987
## LogBook     -0.07280    0.03505  -2.077  0.03783 *
## MinBidShare -1.76665    0.08292 -21.306  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 1407.17  on 699  degrees of freedom
## Residual deviance:  593.76  on 691  degrees of freedom
## AIC: 2529.5
##
## Number of Fisher Scoring iterations: 5
```

Lower P value means higher significance, as is shown, significant covariants are VerifyID and MinBidShare.

(b)

Let's do a Bayesian analysis of the Poisson regression. Let the prior be $\beta \sim \mathcal{N}(0, 100 \cdot (X^\top X)^{-1})$, where X is the $n \times p$ covariate matrix. This is a commonly used prior, which is called Zellner's g-prior. Assume first that the posterior density is approximately multivariate normal:

$$\beta \mid y \sim \mathcal{N}(\tilde{\beta}, J_y^{-1}(\tilde{\beta})),$$

where $\tilde{\beta}$ is the posterior mode and $J_y(\tilde{\beta})$ is the negative Hessian at the posterior mode.

$\tilde{\beta}$ and $J_y(\tilde{\beta})$ can be obtained by numerical optimization (`optim.R`) exactly like you already did for the logistic regression in Lab 2 (but with the log posterior function replaced by the corresponding one for the Poisson model, which you have to code up).

```
y<- matrix(data$nBids, ncol=1)
x<- as.matrix(data[, 2:10])

LogPostPossion <- function(beta,y,X){
  beta <- as.numeric(beta)
  lin_pred <- X %*% beta
  p = length(beta)

  # Poisson log-likelihood
  log_lik <- sum(y * lin_pred - exp(lin_pred))

  # Zellner's g-prior: beta ~ N(0, 100 * (X^T X)^-1)
  log_prior = dmvnorm(beta, rep(0, p), 100*solve(t(X)%*%X), log=TRUE);
  return(log_lik + log_prior)
}

OptimRes <- optim(par = rep(0, ncol(x)),
                  fn = LogPostPossion,
                  X = x, y = y,
                  method = "BFGS",
                  control = list(fnscale = -1, maxit = 1000),
                  hessian = TRUE)
approxPostMode <- matrix(OptimRes$par,1,9)
cov_matrix <- solve(-OptimRes$hessian)# J^{-1}
```

(c)

Let's simulate from the actual posterior of β using the Metropolis algorithm and compare the results with the approximate results in (b). Program a general function that uses the Metropolis algorithm to generate random draws from an arbitrary posterior density. In order to show that it is a general function for any model, we denote the vector of model parameters by θ. Let the proposal density be the multivariate normal density mentioned in Lecture 8 (random walk Metropolis):

$$\theta^{(p)} \mid \theta^{(i-1)} \sim \mathcal{N}(\theta^{(i-1)}, c \cdot \Sigma),$$

where $\Sigma = J_y^{-1}(\tilde{\beta})$ was obtained in (b). The value c is a tuning parameter and should be an input to your Metropolis function. The user of your Metropolis function should be able to supply her own posterior density function, not necessarily for the Poisson regression, and still be able to use your Metropolis function. This is not so straightforward, unless you have come across function objects in R. The note `HowToCodeRWM.pdf` in Lisam describes how you can do this in R.

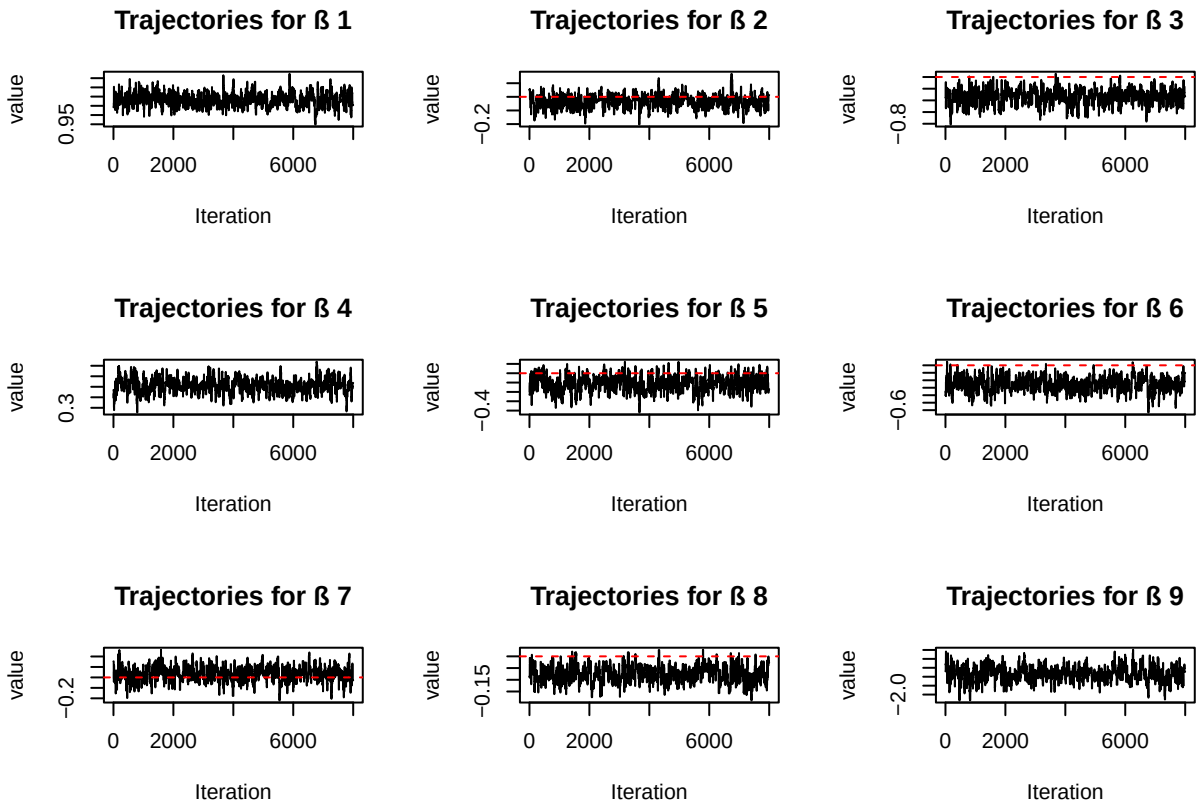
Now, use your new Metropolis function to sample from the posterior of β in the Poisson regression for the eBay dataset. Assess MCMC convergence by graphical methods.

```
mp_sim <- function(c, LogPostFunc, ndraws, theta_init, sigma) {
  p <- length(theta_init)
  theta_mat <- matrix(NA, ndraws, p)
  theta_mat[1, ] <- theta_init

  for (i in 2:ndraws) {
    theta_new <- rmvnorm(1, theta_mat[i - 1, ], c * sigma)

    log_post_new <- LogPostFunc(theta_new, y = y, X = x)
    log_post_old <- LogPostFunc(theta_mat[i - 1, ], y = y, X = x)

    acc_prob <- min(1, exp(log_post_new - log_post_old))

    if (acc_prob > runif(1)) {
      theta_mat[i, ] <- theta_new
    } else {
      theta_mat[i, ] <- theta_mat[i - 1, ]
    }
  }

  return(theta_mat)
}
```

```
set.seed(123)
possion_samp <- mp_sim(1,LogPostFunc = LogPostPossion,
                       ndraws=10000,
                       theta_init=OptimRes$par,
                       sigma=cov_matrix)
```

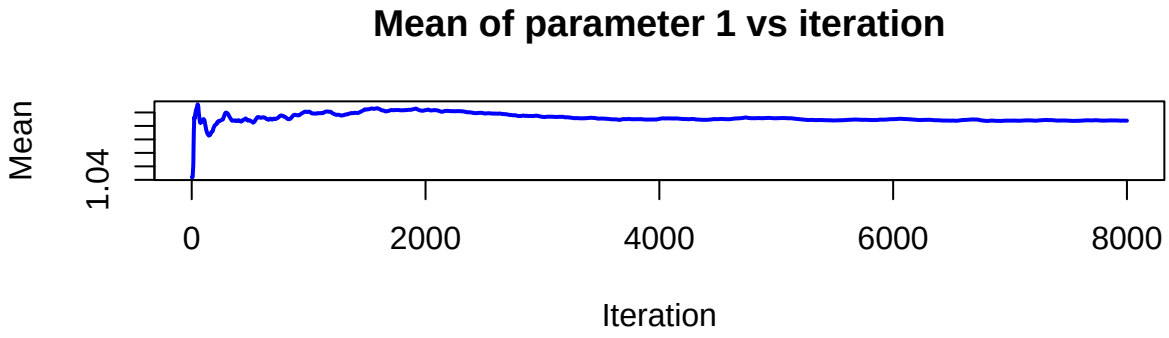
```
par(mfrow = c(3, 3))  # 9 pics
p <- length(OptimRes$par)
burnin <- 2000

for (i in 1:p) {
  plot(
    1:(dim(possion_samp)[1]-burnin),
    possion_samp[(burnin+1):nrow(possion_samp), i],
    type = 'l',
    main = paste("Trajectories for  ", i),
    xlab = "Iteration",
    ylab = "value"
  )
```

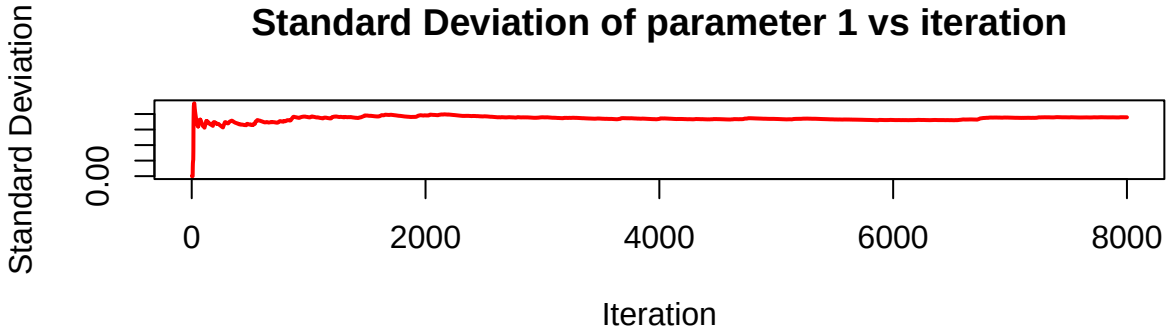
```
  abline(h = 0, col = "red", lty = 2)
}
```

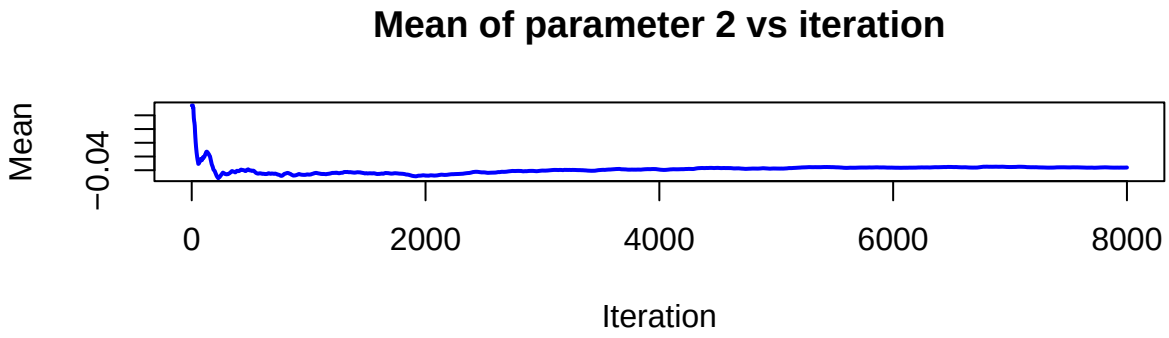


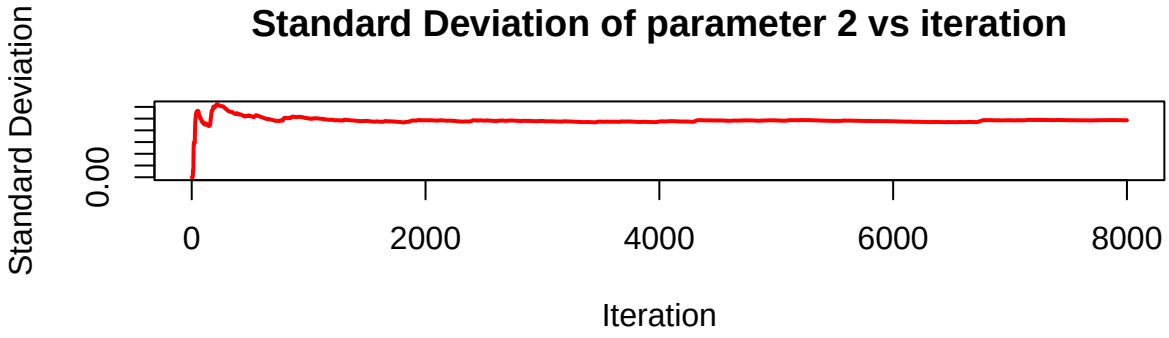

```
ndraws <- nrow(possion_samp)
draw_plot <- ndraws-burnin
for (j in 1:9) {
  param_index <- j  # choose the ith param

mean_vec <- numeric(draw_plot)
sd_vec <- numeric(draw_plot)

for (i in 1:draw_plot) {
  samples_so_far <- possion_samp[(burnin+1):(i+burnin),param_index]
  mean_vec[i] <- mean(samples_so_far)
  sd_vec[i] <- sd(samples_so_far)
}


par(mfrow=c(2,1))

plot(1:draw_plot, mean_vec, type='l', col='blue', lwd=2,
     xlab='Iteration', ylab='Mean',
     main=paste('Mean of parameter', param_index, 'vs iteration'))

plot(1:draw_plot, sd_vec, type='l', col='red', lwd=2,
```

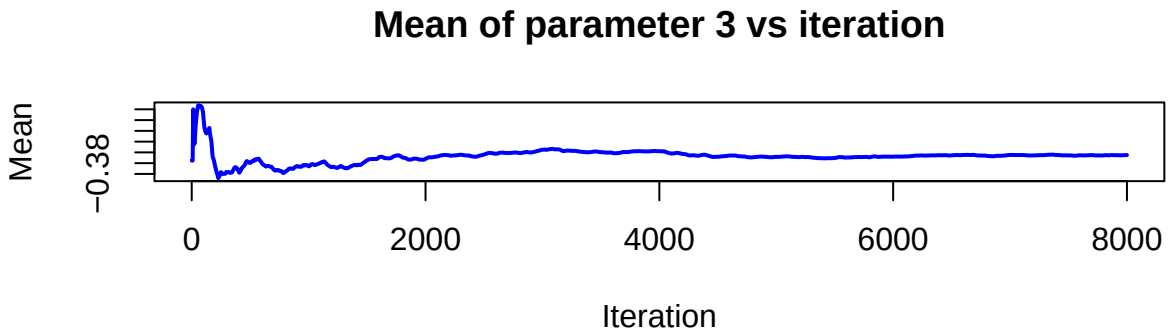
```
    xlab='Iteration', ylab='Standard Deviation',
    main=paste('Standard Deviation of parameter', param_index, 'vs iteration'))
}
```

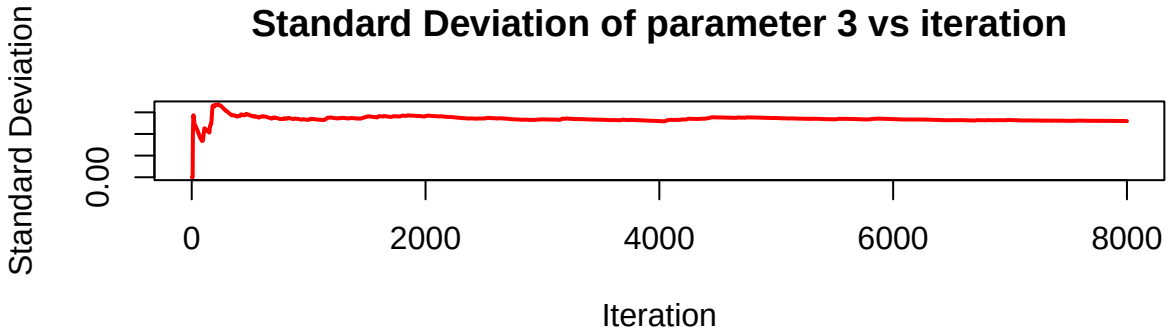


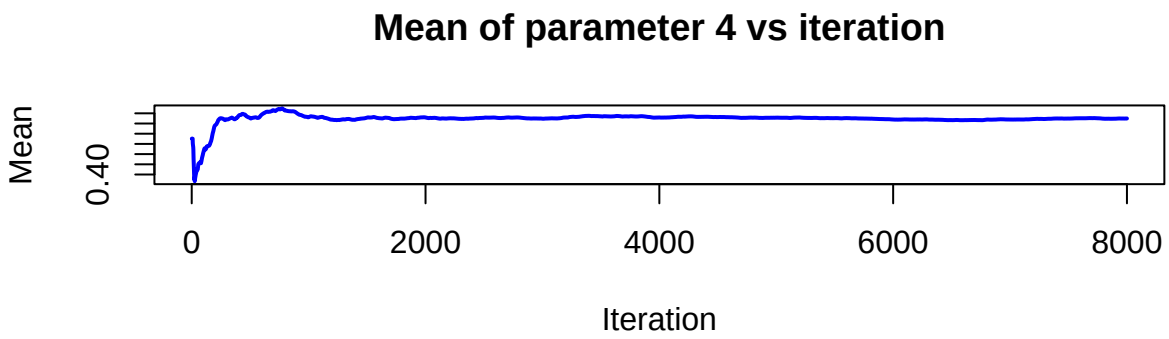


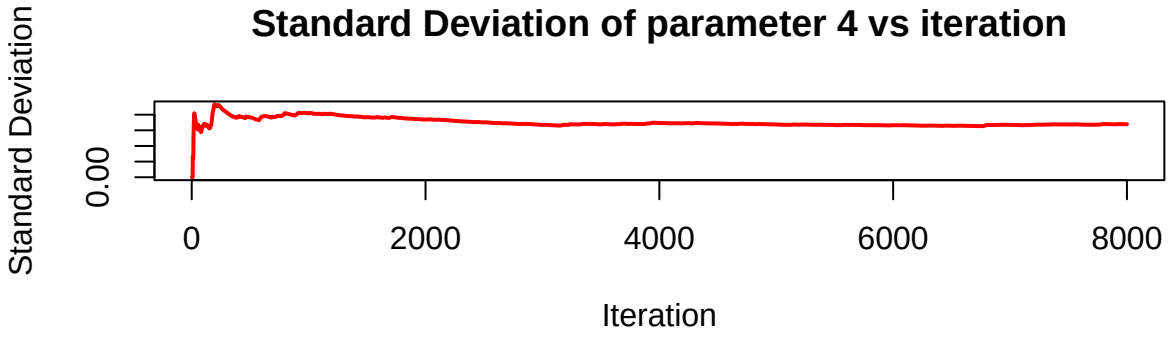

Mean of parameter 2 vs iteration
Mean
−0.04
0
2000
4000
6000
8000
Iteration

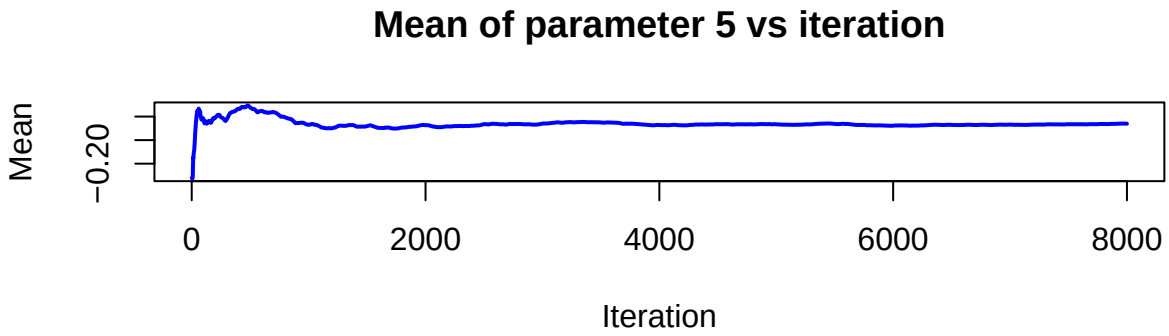

Standard Deviation of parameter 2 vs iteration
Standard Deviation
0.00
0
2000
4000
6000
8000
Iteration

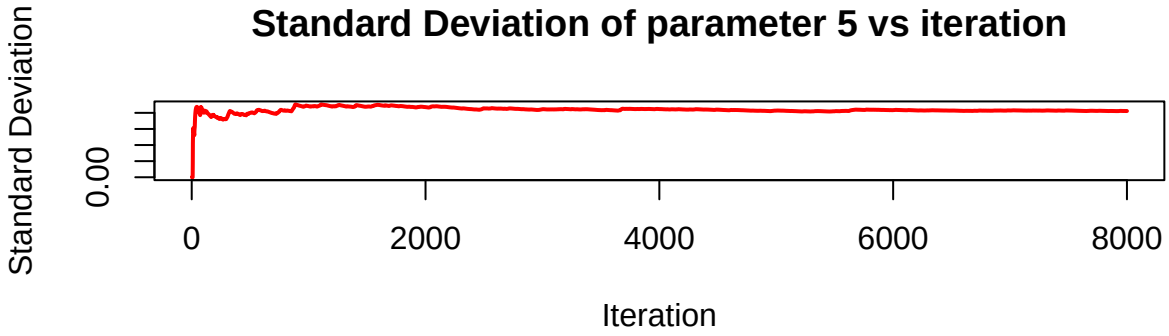

Mean of parameter 3 vs iteration
Mean
−0.38
0
2000
4000
6000
8000
Iteration

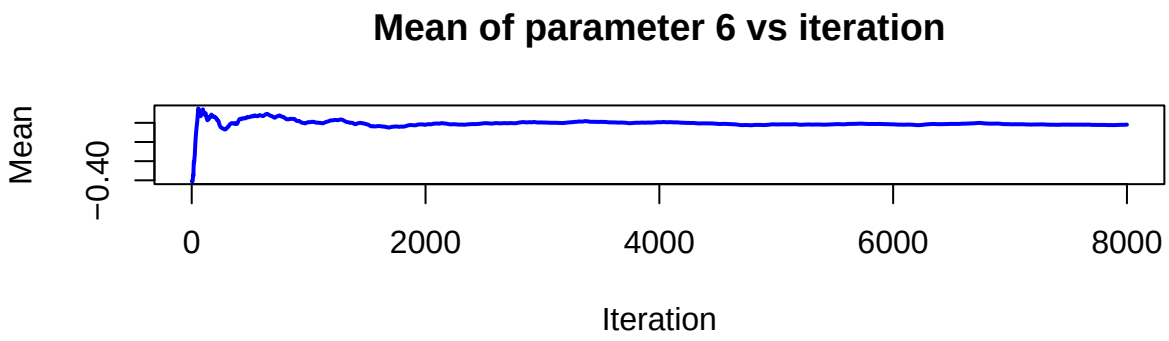

Standard Deviation of parameter 3 vs iteration
Standard Deviation
0.00
0
2000
4000
6000
8000
Iteration

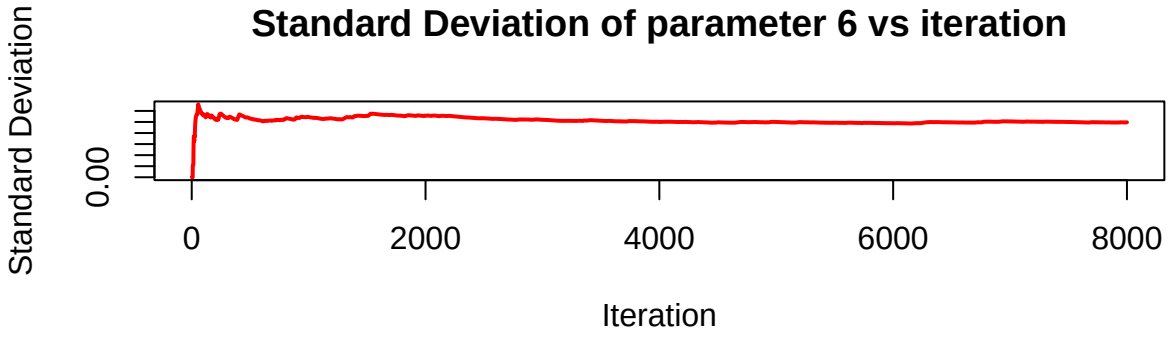

Mean of parameter 4 vs iteration
Mean
0.40
0
2000
4000
6000
8000
Iteration

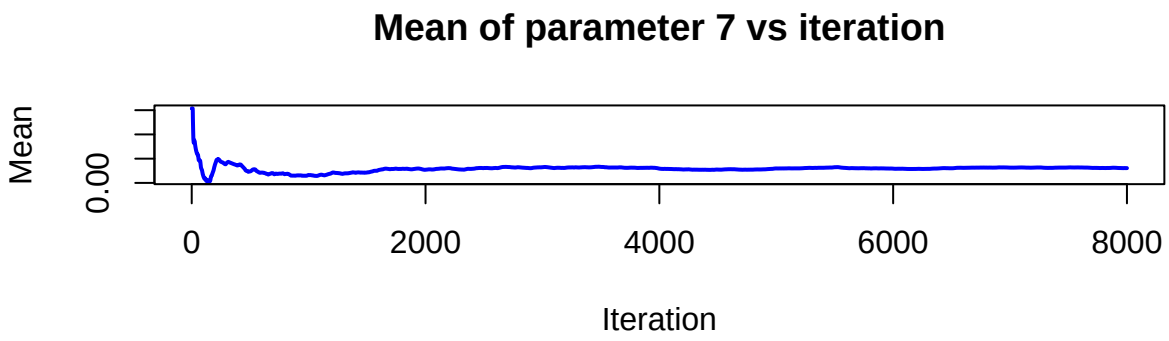

Standard Deviation of parameter 4 vs iteration
Standard Deviation
0.00
0
2000
4000
6000
8000
Iteration

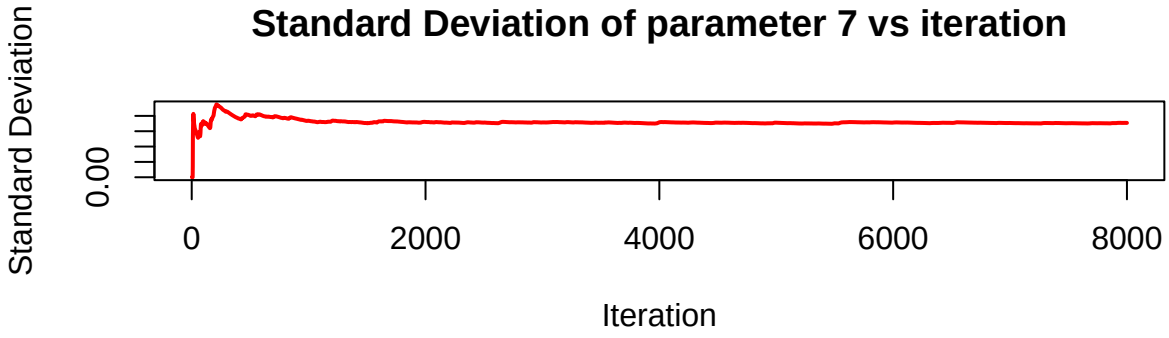

Mean of parameter 5 vs iteration
Mean
−0.20
0
2000
4000
6000
8000
Iteration

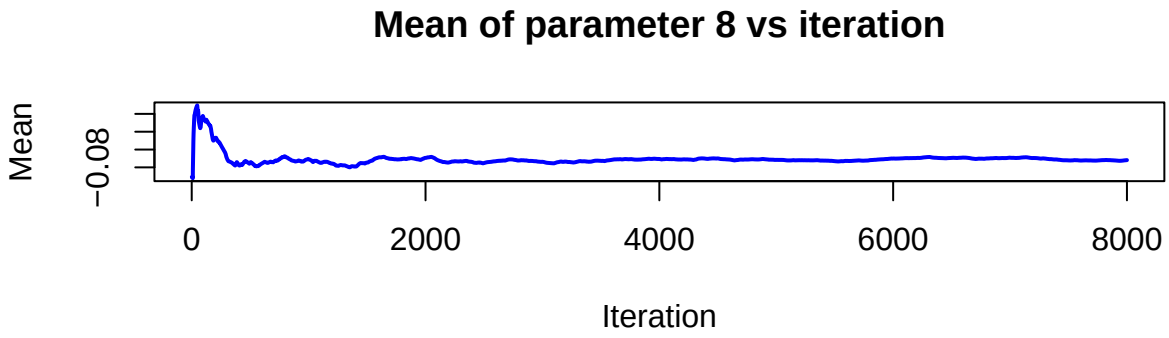

Standard Deviation of parameter 5 vs iteration
Standard Deviation
0.00
0
2000
4000
6000
8000
Iteration

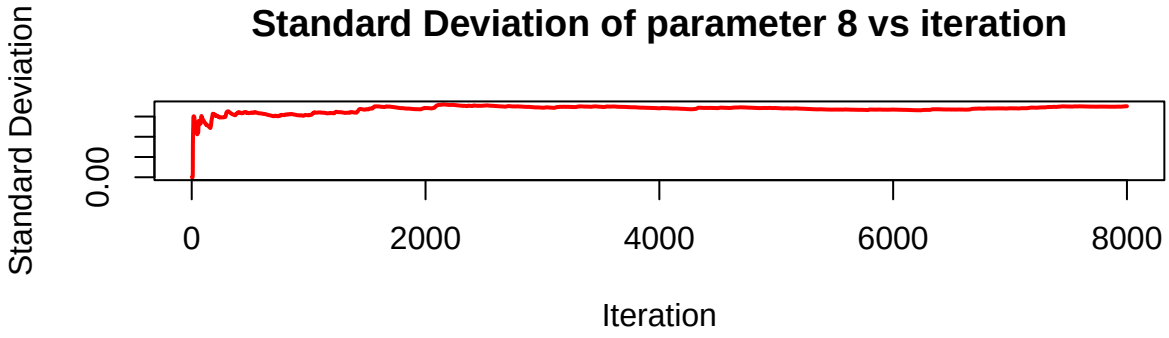

Mean of parameter 6 vs iteration
Mean
−0.40
0
2000
4000
6000
8000
Iteration

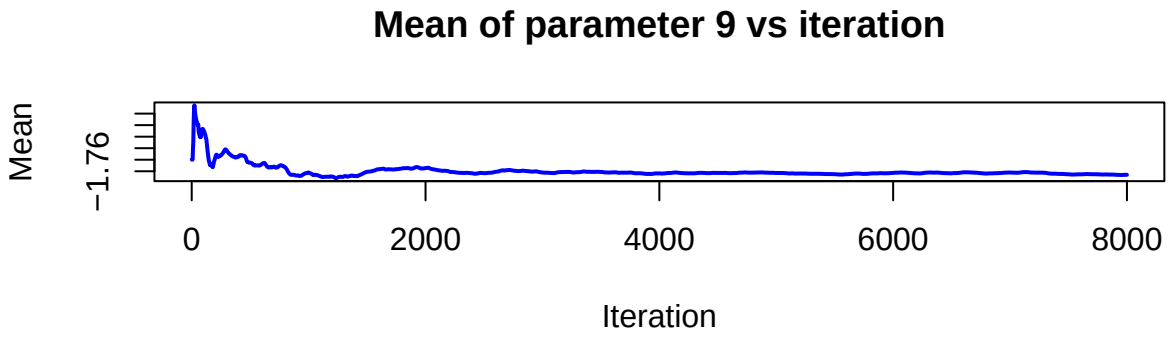

Standard Deviation of parameter 6 vs iteration
Standard Deviation
0.00
0
2000
4000
6000
8000
Iteration

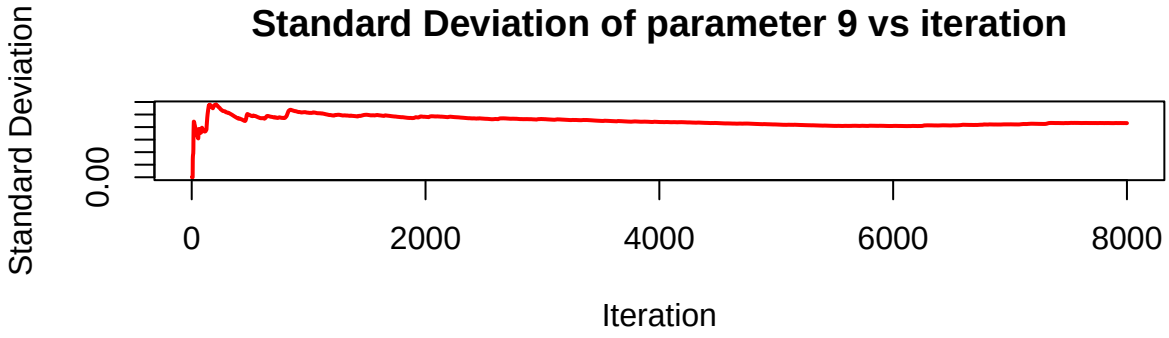

Mean of parameter 7 vs iteration
Mean
0.00
0
2000
4000
6000
8000
Iteration


Standard Deviation of parameter 7 vs iteration
Standard Deviation
0.00
0
2000
4000
6000
8000
Iteration


Mean of parameter 8 vs iteration
Mean
−0.08
0
2000
4000
6000
8000
Iteration


Standard Deviation of parameter 8 vs iteration
Standard Deviation
0.00
0
2000
4000
6000
8000
Iteration





Since trajectories and the mean and standard deviation for different parameters are stable, convergence is thus considered reached.

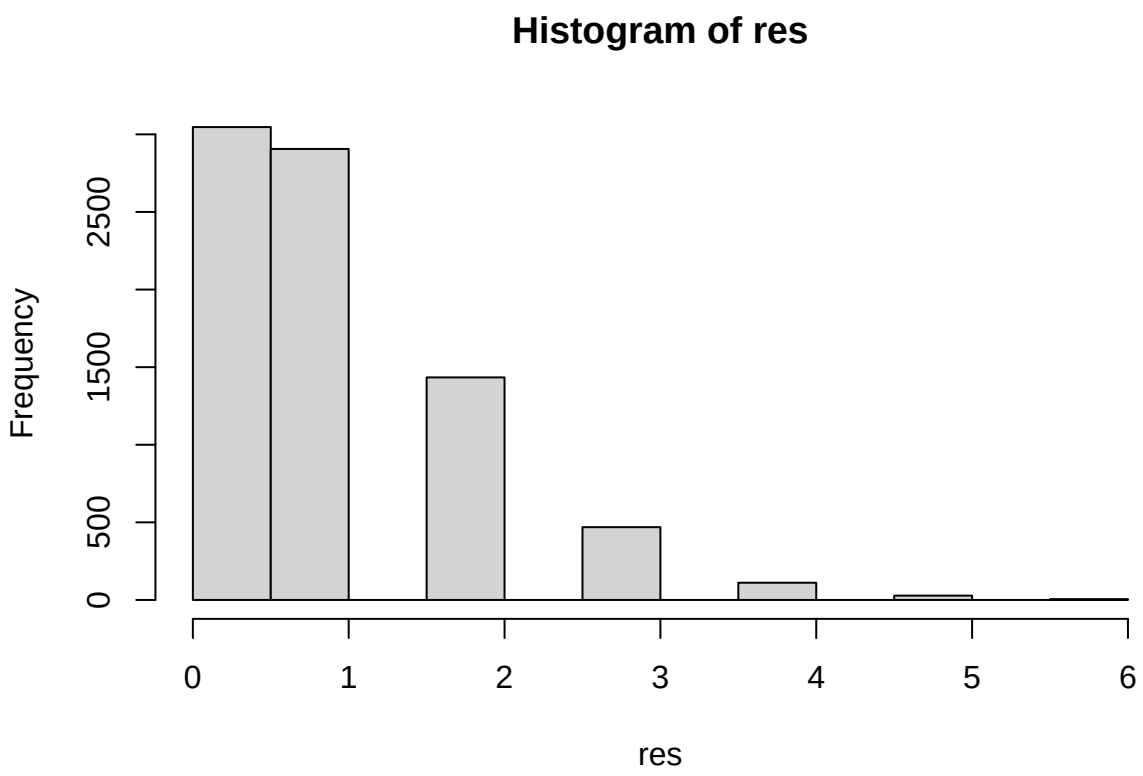
(d)

Use the MCMC draws from (c) to simulate from the predictive distribution of the number of bidders in a new auction with the characteristics below. Plot the predictive distribution. What is the probability of no bidders in this new auction?

- PowerSeller = 1
- VerifyID = 0
- Sealed = 1
- MinBlem = 0
- MajBlem = 1
- LargNeg = 0
- LogBook = 1.3

- MinBidShare = 0.7

```
bidden <- c(1,1,0,1,0,1,0,1.3,0.7) # intercept as 1 to include it
lambda <- exp(possion_samp[(burnin+1):ndraws,] %*% bidden)

res <-rep(NA,length(lambda))
set.seed(124)
for (i in 1:length(lambda)) {
  res[i] <- rpois(1, lambda[i])
}

hist(res)
```




```
prob_0 <- sum(res ==0)/length(lambda)
print(prob_0)
```

```
## [1] 0.380875
```

3.Time series models in Stan

(a)

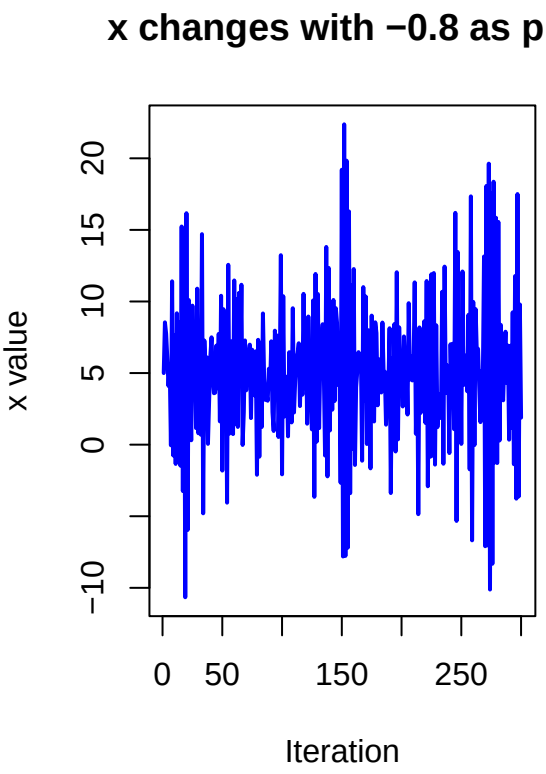
Write a function in R that simulates data from the AR(1)-process

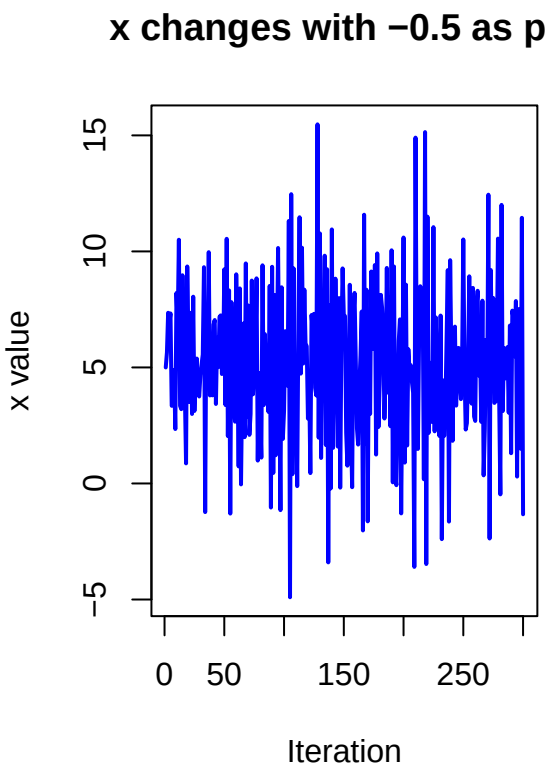
$$x_t = \mu + \phi(x_{t-1} - \mu) + \varepsilon_t, \quad \varepsilon_t \overset{iid}{\sim} \mathcal{N}(0, \sigma^2),$$

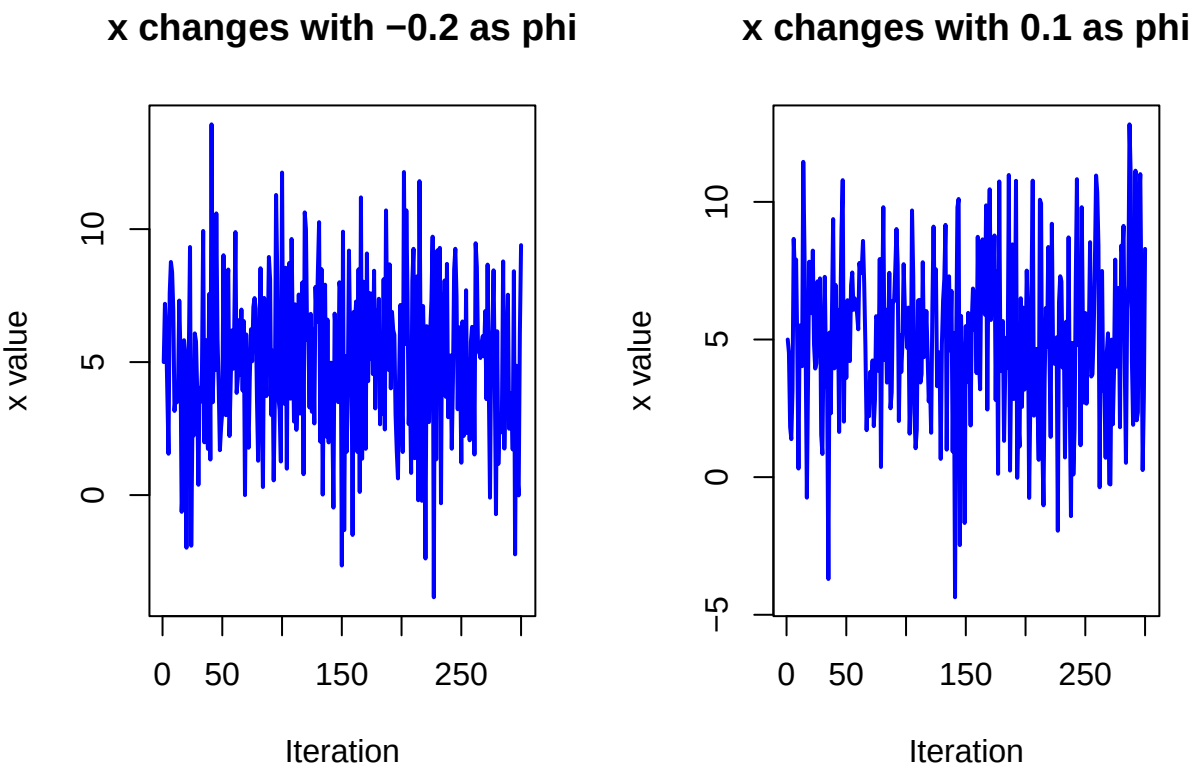
for given values of μ, ϕ and σ^2. Start the process at $x_1 = \mu$ and then simulate values for x_t for $t = 2, 3, \ldots, T$ and return the vector $x_{1:T}$ containing all time points. Use $\mu = 5$, $\sigma^2 = 9$, and $T = 300$ and look at some different realizations (simulations) of $x_{1:T}$ for values of ϕ between -1 and 1 (this is the interval of ϕ where the AR(1)-process is stationary). Include a plot of at least one realization in the report. What effect does the value of ϕ have on $x_{1:T}$?

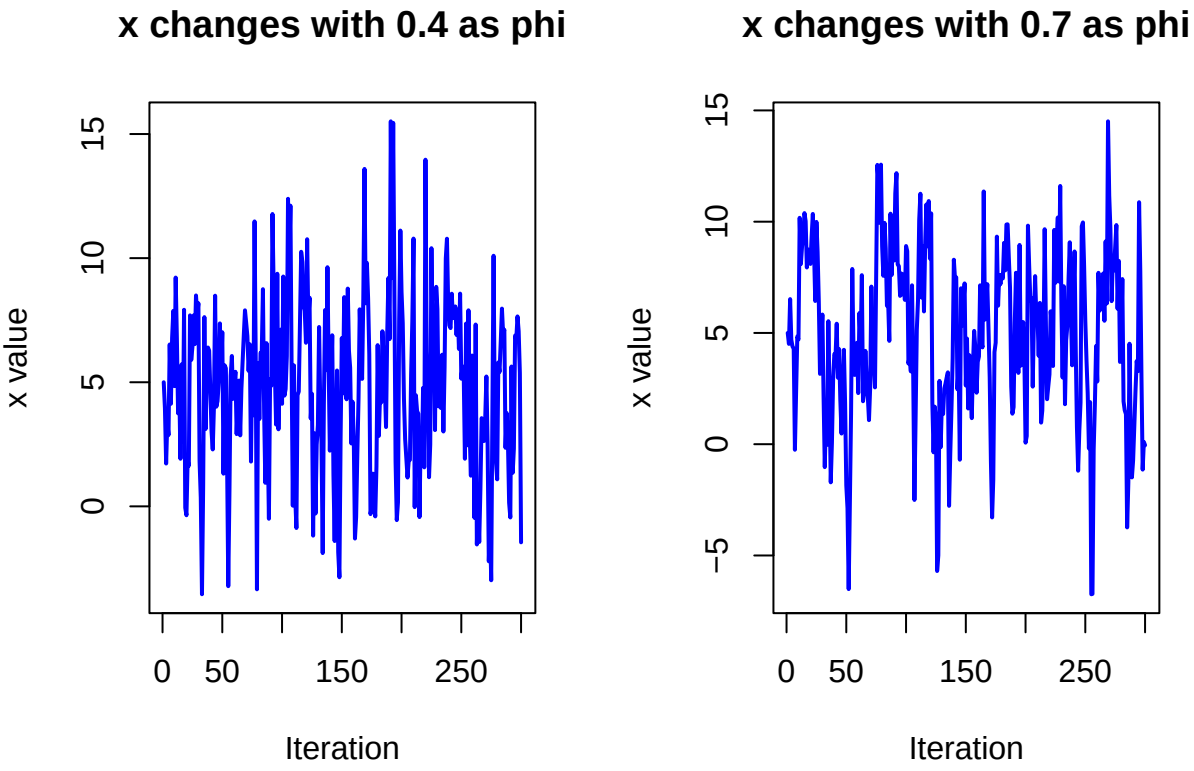
```
rm(list=ls())
#define the function

ar <- function (mu, phi, sig2, T) {
  res<- c()
  x<- mu
  res<-x
  for (t in 2:T) {
    x_new <- mu + phi*(x-mu)+rnorm(1,0,sqrt(sig2))
    res<- c(res,x_new)
    x<- x_new
  }
  return(res)
}

#simulate
set.seed(123)
p <- seq(-1,1,0.1)
res <- matrix(NA,nrow=300, ncol=length(p))
for (i in 1:length(p)) {
  res[,i] <-ar (mu=5, phi=p[i], sig2=9, T=300)

}
```

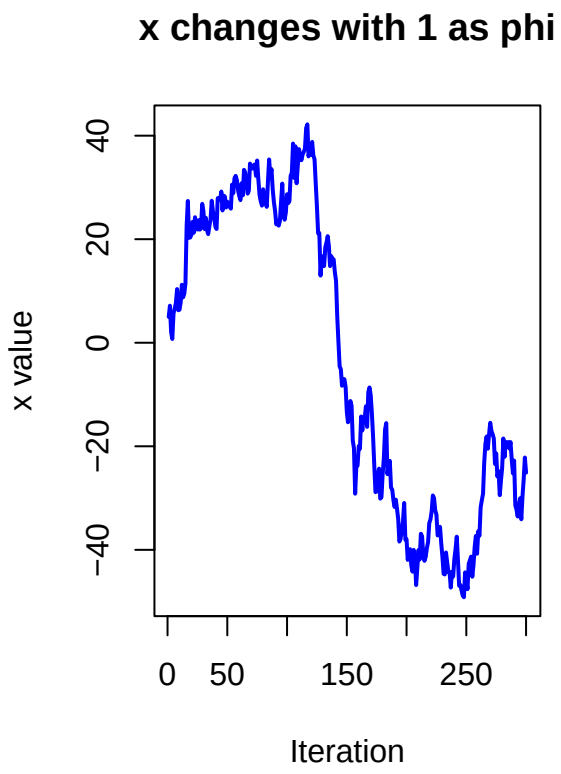
```
nplot <- seq(3,21,3)
par(mfrow = c(1, 2))  # 6 pics
for (i in nplot) {
  plot(res[,i], type='l', col='blue', lwd=2,
     xlab='Iteration', ylab='x value',
     main=paste('x changes with',p[i], 'as phi'))
}
```


x changes with −0.8 as phi
x value
−10
0
5
10
15
20
0
50
150
250
Iteration


x changes with −0.5 as phi
x value
−5
0
5
10
15
0
50
150
250
Iteration


x changes with −0.2 as phi
x value
0
5
10
0
50
150
250
Iteration
x changes with 0.1 as phi
x value
−5
0
5
10
0
50
150
250
Iteration


x changes with 0.4 as phi
x value
0
5
10
15
0
50
150
250
Iteration
x changes with 0.7 as phi
x value
−5
0
5
10
15
0
50
150
250
Iteration



ϕ controls the strength of autocorrelation. ϕ with small absolute value gives weak correlation and values fulctuate quickly.

(b)

Use your function from (a) to simulate two AR(1)-processes, $x_{1:T}$ with $\phi = 0.4$ and $y_{1:T}$ with $\phi = 0.98$. Now, treat your simulated vectors as synthetic data, and treat the values of μ, ϕ and σ^2 as unknown parameters. Implement Stan-code that samples from the posterior of the three parameters, using suitable non-informative priors of your choice.

> *Hint:* Look at the time-series models examples in the Stan user's guide/reference manual, and note the different parameterization used here.

i. Report the posterior mean, 95% credible intervals, and the number of effective posterior samples for the three inferred parameters for each of the simulated AR(1)-processes. Are you able to estimate the true values?

ii. For each of the two data sets, evaluate the convergence of the samplers and plot the joint posterior of μ and ϕ. Comments?

```
set.seed(123)
ar1 <- ar(mu=5, phi=0.4, sig2=9, T=300)
ar2 <- ar(mu=5, phi=0.98, sig2=9, T=300)
```

```r
library(rstan)
```

```
##      StanHeaders
```

```
##
## rstan version 2.32.7 (Stan version 2.32.2)
```

```
## For execution on a local, multicore CPU with excess RAM we recommend calling
## options(mc.cores = parallel::detectCores()).
## To avoid recompilation of unchanged Stan programs, we recommend calling
## rstan_options(auto_write = TRUE)
## For within-chain threading using `reduce_sum()` or `map_rect()` Stan functions,
## change `threads_per_chain` option:
## rstan_options(threads_per_chain = 1)
```

```
## Do not specify '-march=native' in 'LOCAL_CPPFLAGS' or a Makevars file
```

```r
y=ar1
N=length(y)

StanModel = '
data {
  int<lower=0> N; // Number of observations
  vector[N] y; // Number of flowers
}
parameters {
  real mu;
  real<lower=-1, upper=1> phi;
  real<lower=0> sigma;
}
model {
  mu ~ normal(0, 5);
  phi ~ uniform(-1,1);
  sigma ~ normal(3, 3); // Normal with mean 3, st.dev. 3  It is normal (mean, sd_dev) in stan

  for(i in 2:N){
    y[i]~ normal(mu+ phi*(y[i-1] - mu), sigma);
  }
}'



data <- list(N=N, y=y)
warmup <- 50
niter <- 2000
fit <- stan(model_code=StanModel,data=data, warmup=warmup,iter=niter,chains=4)
```

```
## WARNING: Rtools is required to build R packages, but is not currently installed.
##
## Please download and install the appropriate version of Rtools for 4.4.3 from
## https://cran.r-project.org/bin/windows/Rtools/.
```

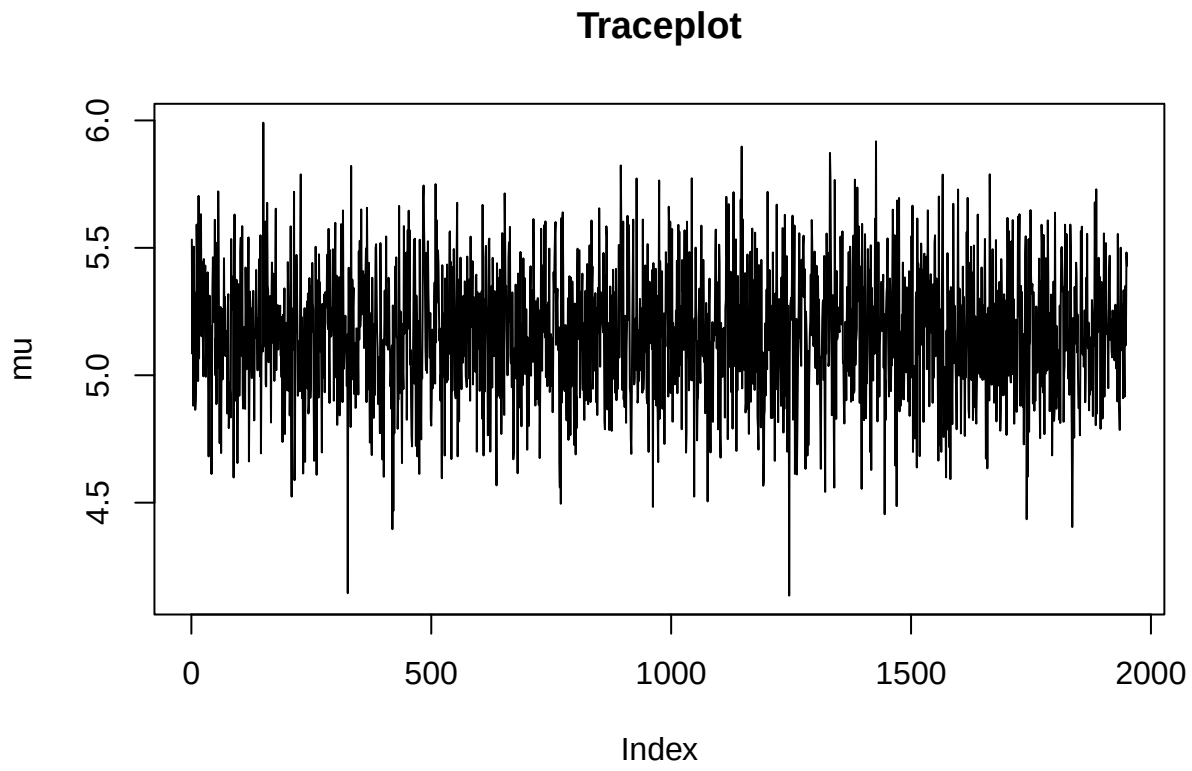
```
## Trying to compile a simple C file

## Running "C:/PROGRA~1/R/R-44~1.3/bin/x64/Rcmd.exe" SHLIB foo.c
## using C compiler: 'gcc.exe (GCC) 13.3.0'
## gcc  -I"C:/PROGRA~1/R/R-44~1.3/include" -DNDEBUG   -I"C:/Users/Administrator/AppData/Local/R/win-
library/4.4/Rcpp/include/"  -I"C:/Users/Administrator/AppData/Local/R/win-library/4.4/RcppEigen/include/
I"C:/Users/Administrator/AppData/Local/R/win-library/4.4/RcppEigen/include/unsupported"  -
I"C:/Users/Administrator/AppData/Local/R/win-library/4.4/BH/include" -I"C:/Users/Administrator/AppData/L
library/4.4/StanHeaders/include/src/"  -I"C:/Users/Administrator/AppData/Local/R/win-
library/4.4/StanHeaders/include/"  -I"C:/Users/Administrator/AppData/Local/R/win-library/4.4/RcppParall
DRCPP_PARALLEL_USE_TBB=1 -I"C:/Users/Administrator/AppData/Local/R/win-library/4.4/rstan/include" -
DEIGEN_NO_DEBUG  -DBOOST_DISABLE_ASSERTS  -DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -
DUSE_STANC3 -DSTRICT_R_HEADERS  -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -
include "C:/Users/Administrator/AppData/Local/R/win-library/4.4/StanHeaders/include/stan/math/prim/fun/L
std=c++1y    -I"C:/rtools44/x86_64-w64-mingw32.static.posix/include"     -O2 -Wall  -
mfpmath=sse -msse2 -mstackrealign  -c foo.c -o foo.o
## cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
## In file included from C:/Users/Administrator/AppData/Local/R/win-library/4.4/RcppEigen/include/Eigen/
##                  from C:/Users/Administrator/AppData/Local/R/win-library/4.4/RcppEigen/include/Eigen/
##                  from C:/Users/Administrator/AppData/Local/R/win-library/4.4/StanHeaders/include/stan
##                  from <command-line>:
## C:/Users/Administrator/AppData/Local/R/win-library/4.4/RcppEigen/include/Eigen/src/Core/util/Macros.h
##   679 | #include <cmath>
##       |          ^~~~~~
## compilation terminated.
## make: *** [C:/PROGRA~1/R/R-44~1.3/etc/x64/Makeconf:289: foo.o] Error 1

## WARNING: Rtools is required to build R packages, but is not currently installed.
##
## Please download and install the appropriate version of Rtools for 4.4.3 from
## https://cran.r-project.org/bin/windows/Rtools/.

##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 1).
## Chain 1:
## Chain 1: Gradient evaluation took 5.8e-05 seconds
## Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 0.58 seconds.
## Chain 1: Adjust your expectations accordingly!
## Chain 1:
## Chain 1:
## Chain 1: WARNING: There aren't enough warmup iterations to fit the
## Chain 1:          three stages of adaptation as currently configured.
## Chain 1:          Reducing each adaptation stage to 15%/75%/10% of
## Chain 1:          the given number of warmup iterations:
## Chain 1:            init_buffer = 7
## Chain 1:            adapt_window = 38
## Chain 1:            term_buffer = 5
## Chain 1:
## Chain 1: Iteration:    1 / 2000 [  0%]  (Warmup)
## Chain 1: Iteration:   51 / 2000 [  2%]  (Sampling)
## Chain 1: Iteration:  250 / 2000 [ 12%]  (Sampling)
## Chain 1: Iteration:  450 / 2000 [ 22%]  (Sampling)
## Chain 1: Iteration:  650 / 2000 [ 32%]  (Sampling)
```

```
## Chain 1: Iteration:  850 / 2000 [ 42%]  (Sampling)
## Chain 1: Iteration: 1050 / 2000 [ 52%]  (Sampling)
## Chain 1: Iteration: 1250 / 2000 [ 62%]  (Sampling)
## Chain 1: Iteration: 1450 / 2000 [ 72%]  (Sampling)
## Chain 1: Iteration: 1650 / 2000 [ 82%]  (Sampling)
## Chain 1: Iteration: 1850 / 2000 [ 92%]  (Sampling)
## Chain 1: Iteration: 2000 / 2000 [100%]  (Sampling)
## Chain 1:
## Chain 1:  Elapsed Time: 0.027 seconds (Warm-up)
## Chain 1:                0.574 seconds (Sampling)
## Chain 1:                0.601 seconds (Total)
## Chain 1:
##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 2).
## Chain 2:
## Chain 2: Gradient evaluation took 3.5e-05 seconds
## Chain 2: 1000 transitions using 10 leapfrog steps per transition would take 0.35 seconds.
## Chain 2: Adjust your expectations accordingly!
## Chain 2:
## Chain 2:
## Chain 2: WARNING: There aren't enough warmup iterations to fit the
## Chain 2:          three stages of adaptation as currently configured.
## Chain 2:          Reducing each adaptation stage to 15%/75%/10% of
## Chain 2:          the given number of warmup iterations:
## Chain 2:            init_buffer = 7
## Chain 2:            adapt_window = 38
## Chain 2:            term_buffer = 5
## Chain 2:
## Chain 2: Iteration:    1 / 2000 [  0%]  (Warmup)
## Chain 2: Iteration:   51 / 2000 [  2%]  (Sampling)
## Chain 2: Iteration:  250 / 2000 [ 12%]  (Sampling)
## Chain 2: Iteration:  450 / 2000 [ 22%]  (Sampling)
## Chain 2: Iteration:  650 / 2000 [ 32%]  (Sampling)
## Chain 2: Iteration:  850 / 2000 [ 42%]  (Sampling)
## Chain 2: Iteration: 1050 / 2000 [ 52%]  (Sampling)
## Chain 2: Iteration: 1250 / 2000 [ 62%]  (Sampling)
## Chain 2: Iteration: 1450 / 2000 [ 72%]  (Sampling)
## Chain 2: Iteration: 1650 / 2000 [ 82%]  (Sampling)
## Chain 2: Iteration: 1850 / 2000 [ 92%]  (Sampling)
## Chain 2: Iteration: 2000 / 2000 [100%]  (Sampling)
## Chain 2:
## Chain 2:  Elapsed Time: 0.029 seconds (Warm-up)
## Chain 2:                0.645 seconds (Sampling)
## Chain 2:                0.674 seconds (Total)
## Chain 2:
##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 3).
## Chain 3:
## Chain 3: Gradient evaluation took 3.1e-05 seconds
## Chain 3: 1000 transitions using 10 leapfrog steps per transition would take 0.31 seconds.
## Chain 3: Adjust your expectations accordingly!
## Chain 3:
## Chain 3:
## Chain 3: WARNING: There aren't enough warmup iterations to fit the
```

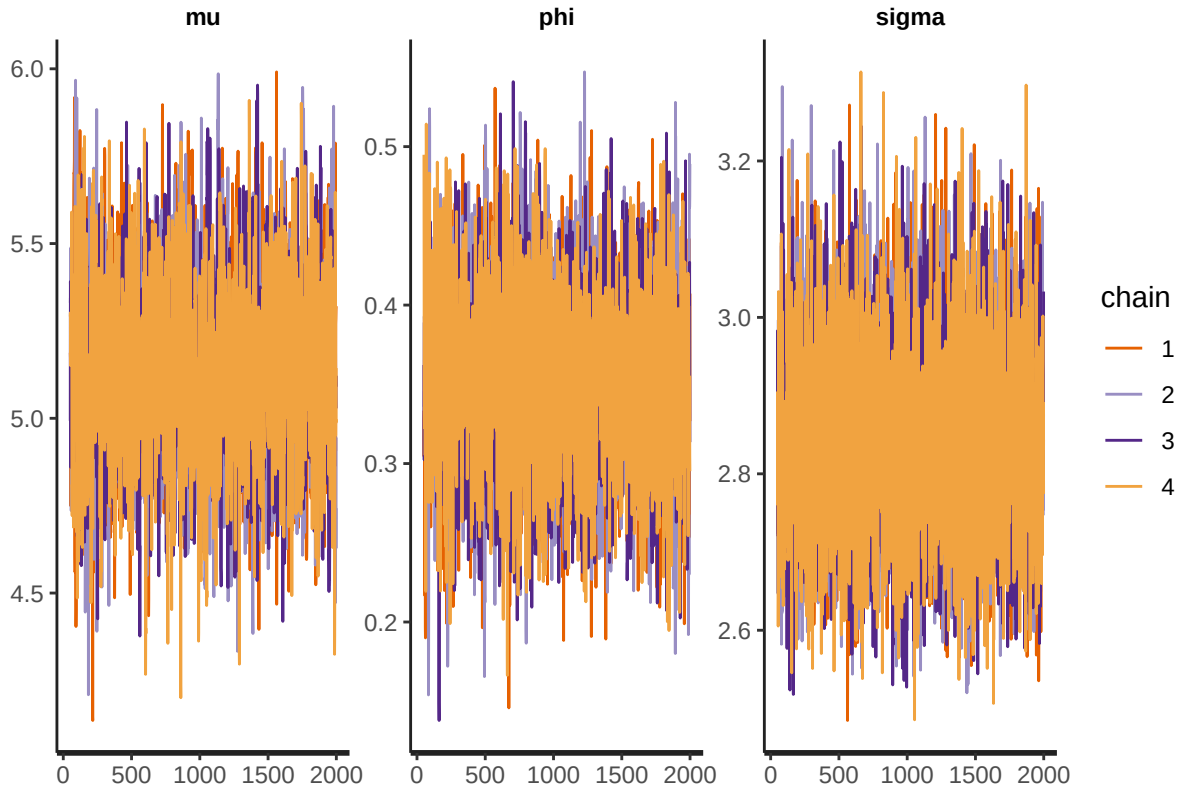
```
## Chain 3:          three stages of adaptation as currently configured.
## Chain 3:          Reducing each adaptation stage to 15%/75%/10% of
## Chain 3:          the given number of warmup iterations:
## Chain 3:            init_buffer = 7
## Chain 3:            adapt_window = 38
## Chain 3:            term_buffer = 5
## Chain 3:
## Chain 3: Iteration:    1 / 2000 [  0%]  (Warmup)
## Chain 3: Iteration:   51 / 2000 [  2%]  (Sampling)
## Chain 3: Iteration:  250 / 2000 [ 12%]  (Sampling)
## Chain 3: Iteration:  450 / 2000 [ 22%]  (Sampling)
## Chain 3: Iteration:  650 / 2000 [ 32%]  (Sampling)
## Chain 3: Iteration:  850 / 2000 [ 42%]  (Sampling)
## Chain 3: Iteration: 1050 / 2000 [ 52%]  (Sampling)
## Chain 3: Iteration: 1250 / 2000 [ 62%]  (Sampling)
## Chain 3: Iteration: 1450 / 2000 [ 72%]  (Sampling)
## Chain 3: Iteration: 1650 / 2000 [ 82%]  (Sampling)
## Chain 3: Iteration: 1850 / 2000 [ 92%]  (Sampling)
## Chain 3: Iteration: 2000 / 2000 [100%]  (Sampling)
## Chain 3:
## Chain 3:  Elapsed Time: 0.017 seconds (Warm-up)
## Chain 3:                0.647 seconds (Sampling)
## Chain 3:                0.664 seconds (Total)
## Chain 3:
##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 4).
## Chain 4:
## Chain 4: Gradient evaluation took 2.7e-05 seconds
## Chain 4: 1000 transitions using 10 leapfrog steps per transition would take 0.27 seconds.
## Chain 4: Adjust your expectations accordingly!
## Chain 4:
## Chain 4:
## Chain 4: WARNING: There aren't enough warmup iterations to fit the
## Chain 4:          three stages of adaptation as currently configured.
## Chain 4:          Reducing each adaptation stage to 15%/75%/10% of
## Chain 4:          the given number of warmup iterations:
## Chain 4:            init_buffer = 7
## Chain 4:            adapt_window = 38
## Chain 4:            term_buffer = 5
## Chain 4:
## Chain 4: Iteration:    1 / 2000 [  0%]  (Warmup)
## Chain 4: Iteration:   51 / 2000 [  2%]  (Sampling)
## Chain 4: Iteration:  250 / 2000 [ 12%]  (Sampling)
## Chain 4: Iteration:  450 / 2000 [ 22%]  (Sampling)
## Chain 4: Iteration:  650 / 2000 [ 32%]  (Sampling)
## Chain 4: Iteration:  850 / 2000 [ 42%]  (Sampling)
## Chain 4: Iteration: 1050 / 2000 [ 52%]  (Sampling)
## Chain 4: Iteration: 1250 / 2000 [ 62%]  (Sampling)
## Chain 4: Iteration: 1450 / 2000 [ 72%]  (Sampling)
## Chain 4: Iteration: 1650 / 2000 [ 82%]  (Sampling)
## Chain 4: Iteration: 1850 / 2000 [ 92%]  (Sampling)
## Chain 4: Iteration: 2000 / 2000 [100%]  (Sampling)
## Chain 4:
## Chain 4:  Elapsed Time: 0.025 seconds (Warm-up)
```

```
## Chain 4:                0.663 seconds (Sampling)
## Chain 4:                0.688 seconds (Total)
## Chain 4:
```

```r
# Print the fitted model
#print(fit,digits_summary=3)
# Extract posterior samples
postDraws <- extract(fit)
# Do traceplots of the first chain
par(mfrow = c(1,1))
plot(postDraws$mu[1:(niter-warmup)],type="l",ylab="mu",main="Traceplot")
```

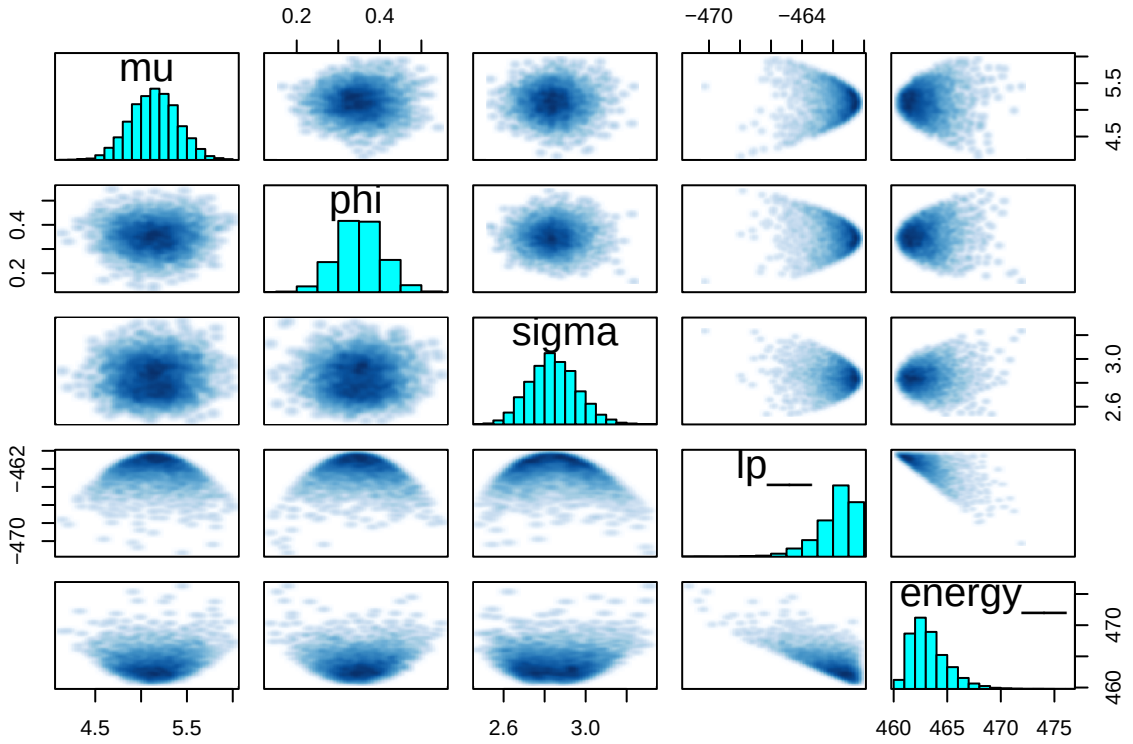



```r
# Do automatic traceplots of all chains
traceplot(fit)
```




```
# Bivariate posterior plots
pairs(fit)
```

```
## Warning in par(usr): argument 1 does not name a graphical parameter
## Warning in par(usr): argument 1 does not name a graphical parameter
## Warning in par(usr): argument 1 does not name a graphical parameter
## Warning in par(usr): argument 1 does not name a graphical parameter
## Warning in par(usr): argument 1 does not name a graphical parameter
```

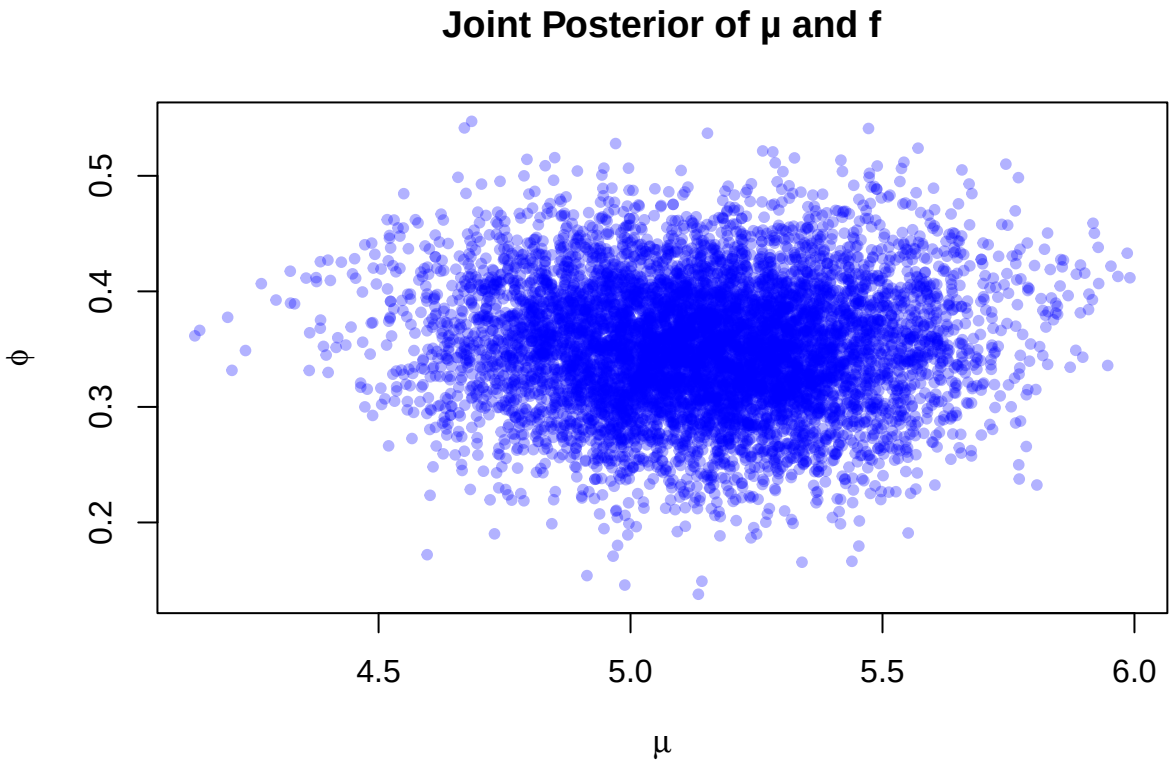



```
summary_fit <- summary(fit, pars = c("mu", "phi", "sigma"))$summary
# extract the mean, 95% CI and effective posterior sample
posterior_summary <- summary_fit[, c("mean", "2.5%", "97.5%", "n_eff" , "Rhat")]
print(round(posterior_summary, 3))
```

```
##         mean  2.5% 97.5%    n_eff  Rhat
## mu    5.144 4.639 5.640 2006.820 1.002
## phi   0.351 0.246 0.459 4176.580 1.001
## sigma 2.844 2.628 3.088 8619.746 1.000
```

Mean, 95% CI and effective posterior sample number are shown as above. The simulated parameter values are quite close to the true parameter values and the true parameter lies in the 95% CI of simulated parameter values, thus we are able to estimate the true values.

```
posterior_samples <- as.data.frame(rstan::extract(fit, pars = c("mu", "phi")))

plot(posterior_samples$mu, posterior_samples$phi,
     xlab = expression(mu), ylab = expression(phi),
     main = "Joint Posterior of   and  ", pch = 20, col = rgb(0, 0, 1, 0.3))
```



```
y=ar2
N=length(y)

StanModel = '
data {
  int<lower=0> N; // Number of observations
  vector[N] y; // Number of flowers
}
parameters {
  real mu;
  real<lower=-1, upper=1> phi;
  real<lower=0> sigma;
}
model {
  mu ~ normal(0, 5);
  phi ~ uniform(-1,1);
  sigma ~ normal(3, 3); // Normal with mean 3, st.dev. 3

  for(i in 2:N){
    y[i]~ normal(mu+ phi*(y[i-1] - mu), sigma);
  }
}'
```

```r
data <- list(N=N, y=y)
warmup <- 50
niter <- 2000
fit <- stan(model_code=StanModel,data=data, warmup=warmup,iter=niter,chains=4)
```

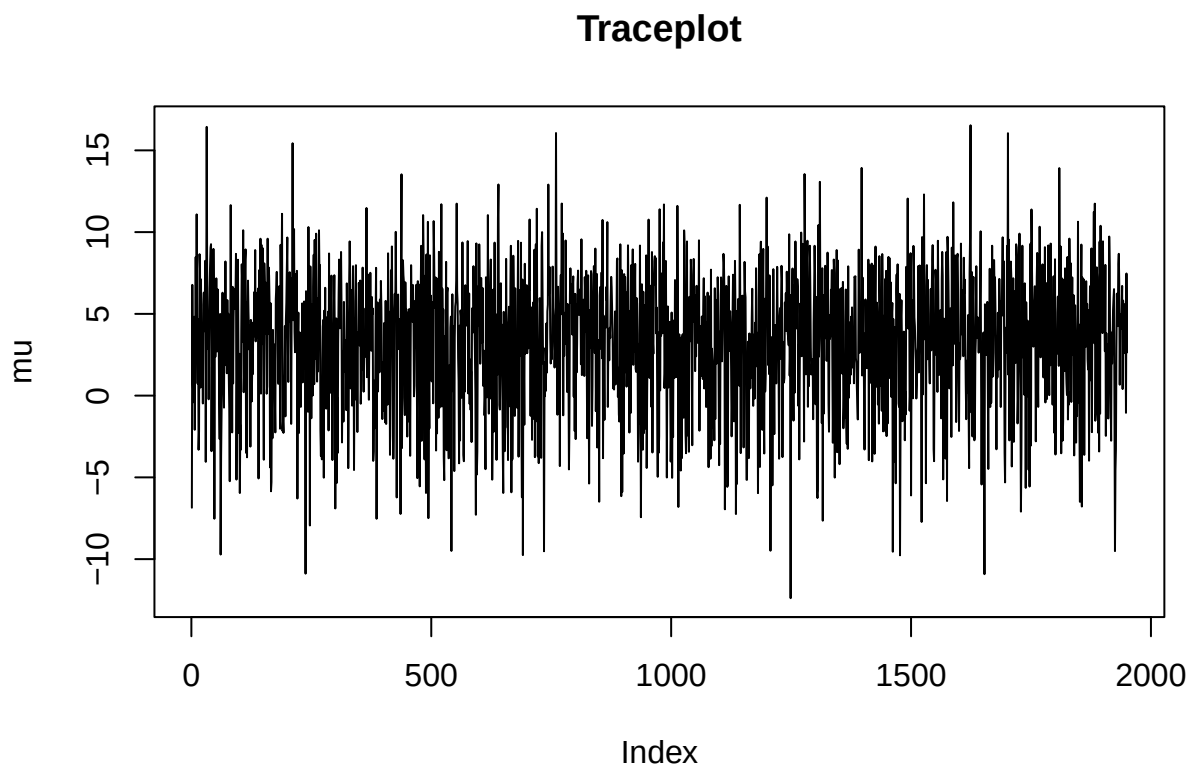
```
## WARNING: Rtools is required to build R packages, but is not currently installed.
##
## Please download and install the appropriate version of Rtools for 4.4.3 from
## https://cran.r-project.org/bin/windows/Rtools/.

## Trying to compile a simple C file

## Running "C:/PROGRA~1/R/R-44~1.3/bin/x64/Rcmd.exe" SHLIB foo.c
## using C compiler: 'gcc.exe (GCC) 13.3.0'
## gcc  -I"C:/PROGRA~1/R/R-44~1.3/include" -DNDEBUG   -I"C:/Users/Administrator/AppData/Local/R/win-
library/4.4/Rcpp/include/"  -I"C:/Users/Administrator/AppData/Local/R/win-library/4.4/RcppEigen/include/
I"C:/Users/Administrator/AppData/Local/R/win-library/4.4/RcppEigen/include/unsupported"  -
I"C:/Users/Administrator/AppData/Local/R/win-library/4.4/BH/include" -I"C:/Users/Administrator/AppData/L
library/4.4/StanHeaders/include/src/"  -I"C:/Users/Administrator/AppData/Local/R/win-
library/4.4/StanHeaders/include/"  -I"C:/Users/Administrator/AppData/Local/R/win-library/4.4/RcppParall
DRCPP_PARALLEL_USE_TBB=1 -I"C:/Users/Administrator/AppData/Local/R/win-library/4.4/rstan/include" -
DEIGEN_NO_DEBUG  -DBOOST_DISABLE_ASSERTS  -DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -
DUSE_STANC3 -DSTRICT_R_HEADERS  -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -
include "C:/Users/Administrator/AppData/Local/R/win-library/4.4/StanHeaders/include/stan/math/prim/fun/L
std=c++1y    -I"C:/rtools44/x86_64-w64-mingw32.static.posix/include"     -O2 -Wall  -
mfpmath=sse -msse2 -mstackrealign  -c foo.c -o foo.o
## cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
## In file included from C:/Users/Administrator/AppData/Local/R/win-library/4.4/RcppEigen/include/Eigen/
##                  from C:/Users/Administrator/AppData/Local/R/win-library/4.4/RcppEigen/include/Eigen/
##                  from C:/Users/Administrator/AppData/Local/R/win-library/4.4/StanHeaders/include/stan
##                  from <command-line>:
## C:/Users/Administrator/AppData/Local/R/win-library/4.4/RcppEigen/include/Eigen/src/Core/util/Macros.h
##   679 | #include <cmath>
##       |          ^~~~~~
## compilation terminated.
## make: *** [C:/PROGRA~1/R/R-44~1.3/etc/x64/Makeconf:289: foo.o] Error 1

## WARNING: Rtools is required to build R packages, but is not currently installed.
##
## Please download and install the appropriate version of Rtools for 4.4.3 from
## https://cran.r-project.org/bin/windows/Rtools/.

##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 1).
## Chain 1:
## Chain 1: Gradient evaluation took 6.6e-05 seconds
## Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 0.66 seconds.
## Chain 1: Adjust your expectations accordingly!
## Chain 1:
## Chain 1:
## Chain 1: WARNING: There aren't enough warmup iterations to fit the
```

```
## Chain 1:          three stages of adaptation as currently configured.
## Chain 1:          Reducing each adaptation stage to 15%/75%/10% of
## Chain 1:          the given number of warmup iterations:
## Chain 1:            init_buffer = 7
## Chain 1:            adapt_window = 38
## Chain 1:            term_buffer = 5
## Chain 1:
## Chain 1: Iteration:    1 / 2000 [  0%]  (Warmup)
## Chain 1: Iteration:   51 / 2000 [  2%]  (Sampling)
## Chain 1: Iteration:  250 / 2000 [ 12%]  (Sampling)
## Chain 1: Iteration:  450 / 2000 [ 22%]  (Sampling)
## Chain 1: Iteration:  650 / 2000 [ 32%]  (Sampling)
## Chain 1: Iteration:  850 / 2000 [ 42%]  (Sampling)
## Chain 1: Iteration: 1050 / 2000 [ 52%]  (Sampling)
## Chain 1: Iteration: 1250 / 2000 [ 62%]  (Sampling)
## Chain 1: Iteration: 1450 / 2000 [ 72%]  (Sampling)
## Chain 1: Iteration: 1650 / 2000 [ 82%]  (Sampling)
## Chain 1: Iteration: 1850 / 2000 [ 92%]  (Sampling)
## Chain 1: Iteration: 2000 / 2000 [100%]  (Sampling)
## Chain 1:
## Chain 1:  Elapsed Time: 0.105 seconds (Warm-up)
## Chain 1:                4.179 seconds (Sampling)
## Chain 1:                4.284 seconds (Total)
## Chain 1:
##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 2).
## Chain 2:
## Chain 2: Gradient evaluation took 2.8e-05 seconds
## Chain 2: 1000 transitions using 10 leapfrog steps per transition would take 0.28 seconds.
## Chain 2: Adjust your expectations accordingly!
## Chain 2:
## Chain 2:
## Chain 2: WARNING: There aren't enough warmup iterations to fit the
## Chain 2:          three stages of adaptation as currently configured.
## Chain 2:          Reducing each adaptation stage to 15%/75%/10% of
## Chain 2:          the given number of warmup iterations:
## Chain 2:            init_buffer = 7
## Chain 2:            adapt_window = 38
## Chain 2:            term_buffer = 5
## Chain 2:
## Chain 2: Iteration:    1 / 2000 [  0%]  (Warmup)
## Chain 2: Iteration:   51 / 2000 [  2%]  (Sampling)
## Chain 2: Iteration:  250 / 2000 [ 12%]  (Sampling)
## Chain 2: Iteration:  450 / 2000 [ 22%]  (Sampling)
## Chain 2: Iteration:  650 / 2000 [ 32%]  (Sampling)
## Chain 2: Iteration:  850 / 2000 [ 42%]  (Sampling)
## Chain 2: Iteration: 1050 / 2000 [ 52%]  (Sampling)
## Chain 2: Iteration: 1250 / 2000 [ 62%]  (Sampling)
## Chain 2: Iteration: 1450 / 2000 [ 72%]  (Sampling)
## Chain 2: Iteration: 1650 / 2000 [ 82%]  (Sampling)
## Chain 2: Iteration: 1850 / 2000 [ 92%]  (Sampling)
## Chain 2: Iteration: 2000 / 2000 [100%]  (Sampling)
## Chain 2:
## Chain 2:  Elapsed Time: 0.061 seconds (Warm-up)
```

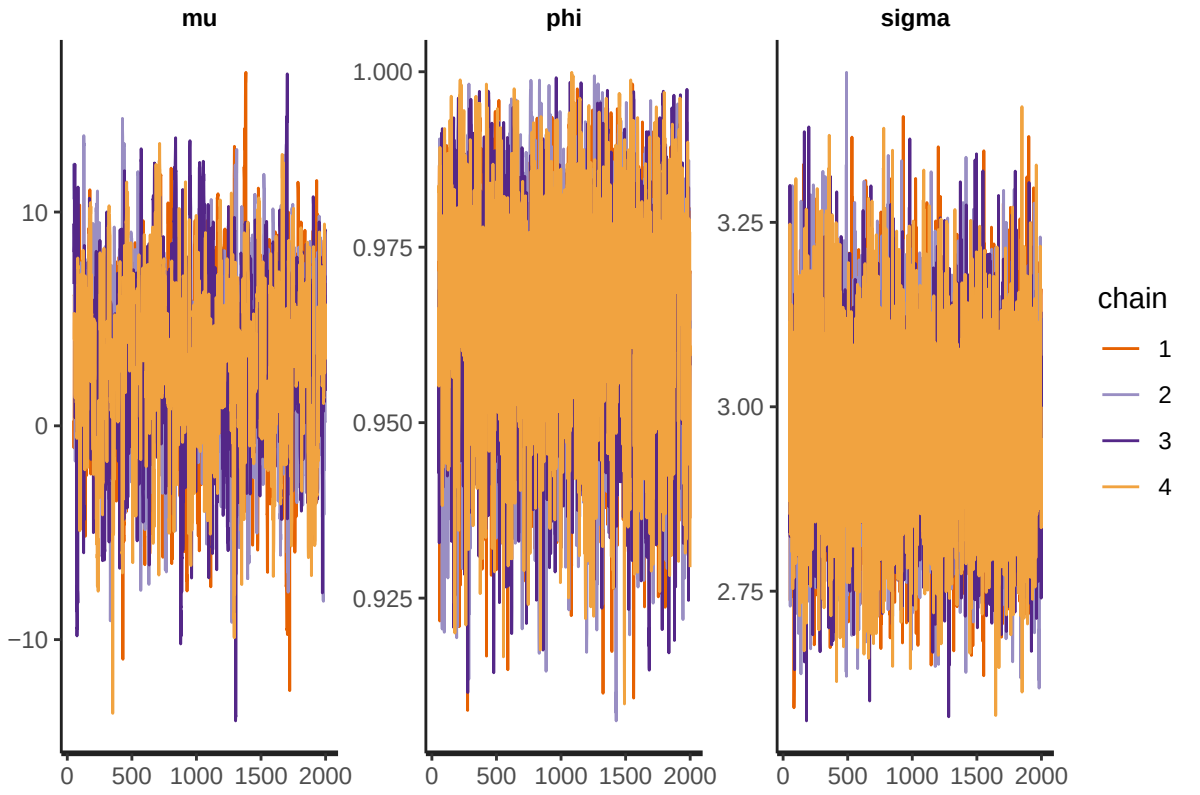
```
## Chain 2:                3.975 seconds (Sampling)
## Chain 2:                4.036 seconds (Total)
## Chain 2:
##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 3).
## Chain 3:
## Chain 3: Gradient evaluation took 2.8e-05 seconds
## Chain 3: 1000 transitions using 10 leapfrog steps per transition would take 0.28 seconds.
## Chain 3: Adjust your expectations accordingly!
## Chain 3:
## Chain 3:
## Chain 3: WARNING: There aren't enough warmup iterations to fit the
## Chain 3:          three stages of adaptation as currently configured.
## Chain 3:          Reducing each adaptation stage to 15%/75%/10% of
## Chain 3:          the given number of warmup iterations:
## Chain 3:            init_buffer = 7
## Chain 3:            adapt_window = 38
## Chain 3:            term_buffer = 5
## Chain 3:
## Chain 3: Iteration:    1 / 2000 [  0%]  (Warmup)
## Chain 3: Iteration:   51 / 2000 [  2%]  (Sampling)
## Chain 3: Iteration:  250 / 2000 [ 12%]  (Sampling)
## Chain 3: Iteration:  450 / 2000 [ 22%]  (Sampling)
## Chain 3: Iteration:  650 / 2000 [ 32%]  (Sampling)
## Chain 3: Iteration:  850 / 2000 [ 42%]  (Sampling)
## Chain 3: Iteration: 1050 / 2000 [ 52%]  (Sampling)
## Chain 3: Iteration: 1250 / 2000 [ 62%]  (Sampling)
## Chain 3: Iteration: 1450 / 2000 [ 72%]  (Sampling)
## Chain 3: Iteration: 1650 / 2000 [ 82%]  (Sampling)
## Chain 3: Iteration: 1850 / 2000 [ 92%]  (Sampling)
## Chain 3: Iteration: 2000 / 2000 [100%]  (Sampling)
## Chain 3:
## Chain 3:  Elapsed Time: 0.1 seconds (Warm-up)
## Chain 3:                4.154 seconds (Sampling)
## Chain 3:                4.254 seconds (Total)
## Chain 3:
##
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 4).
## Chain 4:
## Chain 4: Gradient evaluation took 2.8e-05 seconds
## Chain 4: 1000 transitions using 10 leapfrog steps per transition would take 0.28 seconds.
## Chain 4: Adjust your expectations accordingly!
## Chain 4:
## Chain 4:
## Chain 4: WARNING: There aren't enough warmup iterations to fit the
## Chain 4:          three stages of adaptation as currently configured.
## Chain 4:          Reducing each adaptation stage to 15%/75%/10% of
## Chain 4:          the given number of warmup iterations:
## Chain 4:            init_buffer = 7
## Chain 4:            adapt_window = 38
## Chain 4:            term_buffer = 5
## Chain 4:
## Chain 4: Iteration:    1 / 2000 [  0%]  (Warmup)
## Chain 4: Iteration:   51 / 2000 [  2%]  (Sampling)
```

```
## Chain 4: Iteration:  250 / 2000 [ 12%]  (Sampling)
## Chain 4: Iteration:  450 / 2000 [ 22%]  (Sampling)
## Chain 4: Iteration:  650 / 2000 [ 32%]  (Sampling)
## Chain 4: Iteration:  850 / 2000 [ 42%]  (Sampling)
## Chain 4: Iteration: 1050 / 2000 [ 52%]  (Sampling)
## Chain 4: Iteration: 1250 / 2000 [ 62%]  (Sampling)
## Chain 4: Iteration: 1450 / 2000 [ 72%]  (Sampling)
## Chain 4: Iteration: 1650 / 2000 [ 82%]  (Sampling)
## Chain 4: Iteration: 1850 / 2000 [ 92%]  (Sampling)
## Chain 4: Iteration: 2000 / 2000 [100%]  (Sampling)
## Chain 4:
## Chain 4:  Elapsed Time: 0.051 seconds (Warm-up)
## Chain 4:                3.682 seconds (Sampling)
## Chain 4:                3.733 seconds (Total)
## Chain 4:
```

```r
# Print the fitted model
#print(fit,digits_summary=3)
# Extract posterior samples
postDraws <- extract(fit)
# Do traceplots of the first chain
par(mfrow = c(1,1))
plot(postDraws$mu[1:(niter-warmup)],type="l",ylab="mu",main="Traceplot")
```

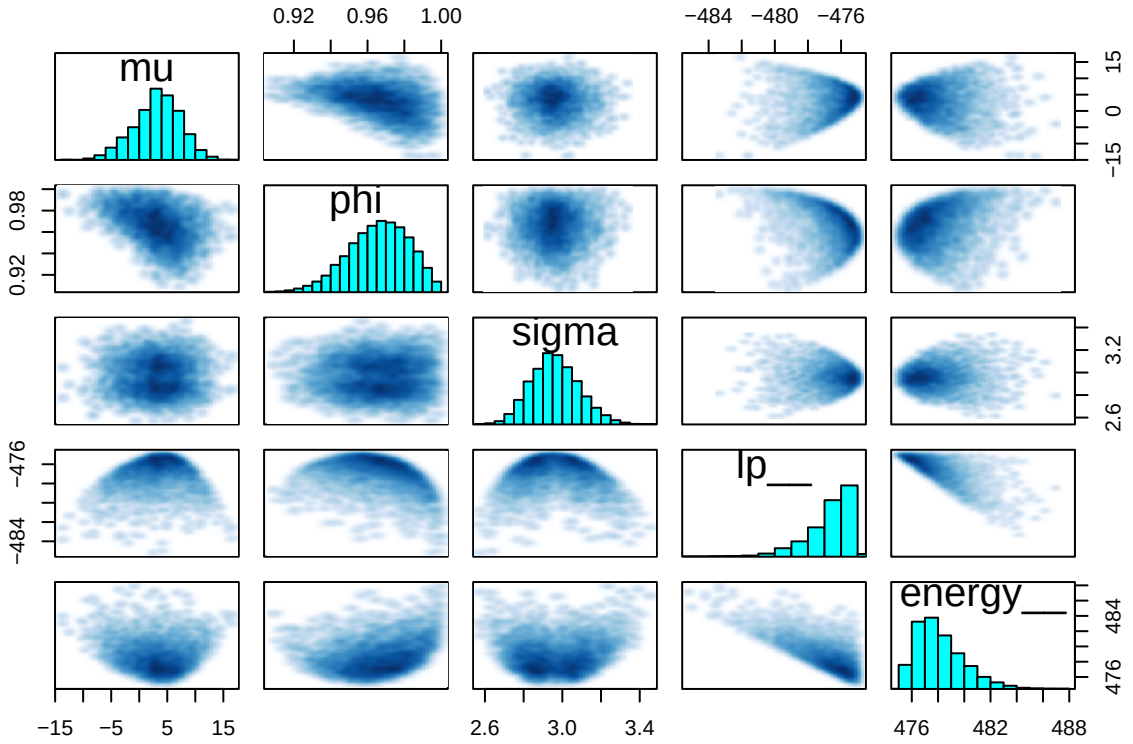


```
# Do automatic traceplots of all chains
traceplot(fit)
```




```
# Bivariate posterior plots
pairs(fit)
```

```
## Warning in par(usr): argument 1 does not name a graphical parameter
## Warning in par(usr): argument 1 does not name a graphical parameter
## Warning in par(usr): argument 1 does not name a graphical parameter
## Warning in par(usr): argument 1 does not name a graphical parameter
## Warning in par(usr): argument 1 does not name a graphical parameter
```

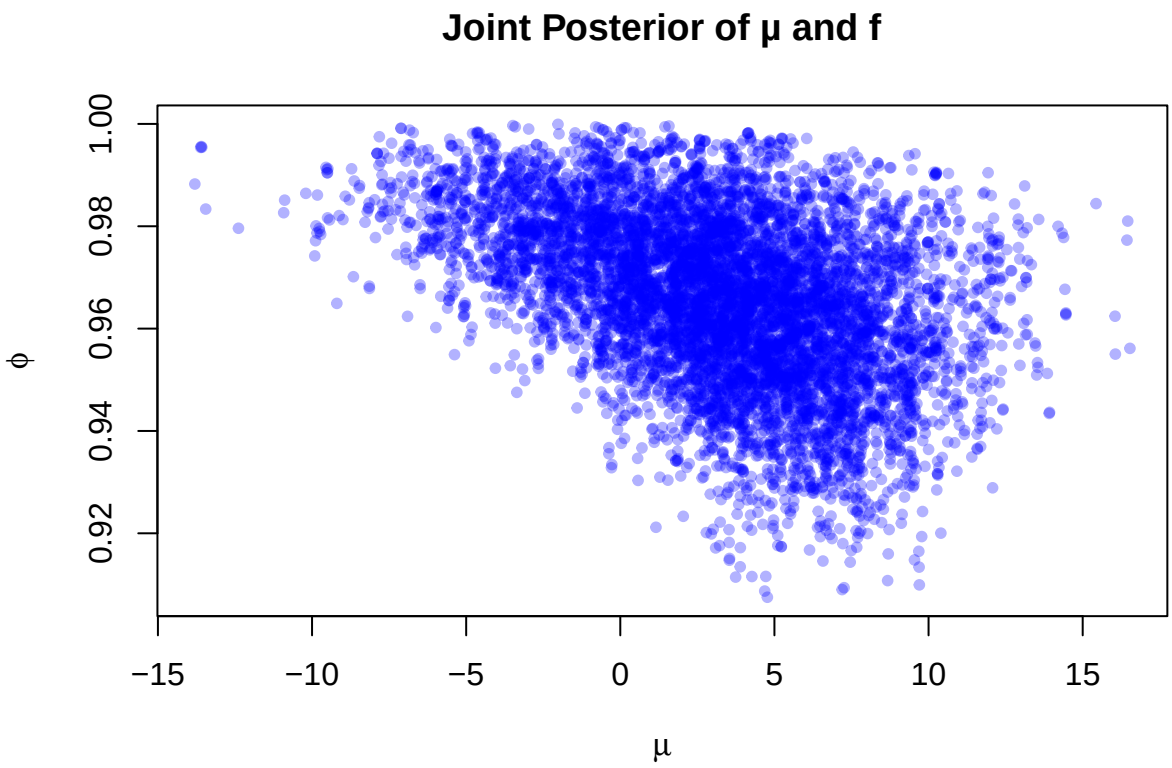



```
summary_fit <- summary(fit, pars = c("mu", "phi", "sigma"))$summary
# extract the mean, 95% CI and effective posterior sample
posterior_summary <- summary_fit[, c("mean", "2.5%", "97.5%", "n_eff", "Rhat")]
print(round(posterior_summary, 3))
```

```
##         mean   2.5%  97.5%    n_eff  Rhat
## mu    3.191 -5.583 10.723  951.075 1.003
## phi   0.966  0.931  0.994 1896.228 1.001
## sigma 2.963  2.739  3.218 7729.057 1.000
```

Mean, 95% CI and effective posterior sample number are shown as above. Despite simulated phi and sigma values are quite close to the true values and the true values lie in the 95% CI, the simulated mean gives too wide 95% CI, making it unable to make a estimate of it. This shows that the data provide little information about , and the estimate of is not very precise.

```
posterior_samples <- as.data.frame(rstan::extract(fit, pars = c("mu", "phi")))

plot(posterior_samples$mu, posterior_samples$phi,
     xlab = expression(mu), ylab = expression(phi),
     main = "Joint Posterior of  and ", pch = 20, col = rgb(0, 0, 1, 0.3))
```



Both cases have Rhat values < 1.05, meaning all chains are well- mixed and agree. Besides, the traceplots have shown no stickiness and no drift. Thus, convergence is considered reached for both ϕ values.
When ϕ is equal to 0.4, μ and ϕ is low correlated (cloud-shape plot) and moderate correlated (diagonal elliptical shape) when ϕ is 0.98. This is due to the near-nonstationary behavior of the AR(1) process when ϕ is close to 1. We can conclude that even if convergence is good or acceptable, it does not necessarily mean a good sample for the posterior.